

FastSAR: High-Throughput, High-Fidelity Open-Source SAR Image Formation on CPUs, GPUs and TPUs

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Abstract

Exact backprojection of a spaceborne SAR collection needs one interpolated read per pixel and pulse, 1.6 trillion for a 12,207 by 8,808 pixel Umbra image of the Panama Canal, and ISCE3, the fastest open-source backprojection that forms this image, would take an estimated 4.8 h on a 16-vCPU instance. FastSAR, an open-source library with backprojection kernels for CPUs, Nvidia GPUs and Google TPUs, forms the image by factorized backprojection in 11.3 s on that instance, 3.8 s on an Nvidia L4 and 3.1 s on a Google TPU v6e. Its errors in three regions of the image are -56 to -61 dB relative to a float64 exact backprojection, and FastSAR’s own exact backprojection, which takes 19 s on the L4, lowers the whole-image error by more than 10 dB. At float32-class accuracy a thousand images cost \$0.75 on the L4, \$1.32 to \$1.58 on the TPUs and \$3.04 on the CPU. On five Capella collections, including a sliding spotlight and two stripmaps, FastSAR’s worst-region error is -59.1 to -61.5 dB, 10 to 34 dB below ISCE3’s when both are scored with the per-pulse antenna positions of the phase-history files.

1 Introduction

Image formation turns a SAR phase history of 10^8 to 10^9 complex samples into an image of similar size. Exact backprojection makes no approximation to the imaging geometry [1, 2], but it needs one interpolated read per pixel and pulse, 1.6 trillion for the Panama image used here (the product of the pixel and pulse counts of Table 1). RITSAR, the AFRL toolbox and ISCE3, whose backprojections produce an image of this collection, would take an estimated 4.8 to 69 h on a 16-vCPU instance (Table 3).

FastSAR is an open-source SAR image formation library with one Python interface for CPUs, Nvidia GPUs and Google TPUs (Section 3). On the same CPU instance and on an Nvidia L4 GPU, the estimated times of those open-source backprojections are 430 to 22,000 times FastSAR’s time on the Panama image (Table 3). The errors of FastSAR’s float32-class and exact images are lower in every region scored, and the estimates are upper bounds (Section 5.1). The smallest ratios, about 1,500 on the CPU and 430 on the L4, are those against ISCE3.

The main algorithm, factorized backprojection, is written as tile-wise phase rotations and decimation filters followed by dense matrix products, with a kernel for each device (Appendix D). Tile geometry is computed in float64, except at the last level of the TPU kernels and of the JAX program (Appendix A). Exact backprojection has CPU and CUDA kernels of its own and is the more accurate algorithm, but on Panama it is 36 times slower than the factorized algorithm on the CPU and 4.9 times slower on the L4 (Table 4).

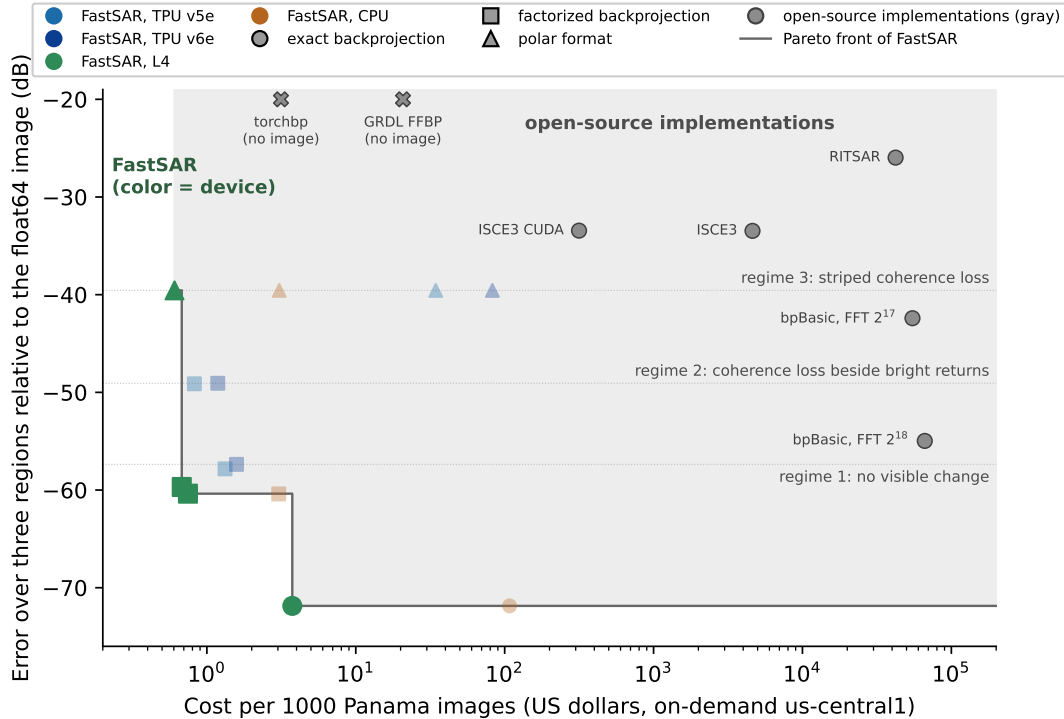


Figure 1: Cost per thousand Panama Canal images against the error over the lock, port and ship regions of Figure 3 (pooled by reference energy, relative to the float64 reference), for FastSAR on each device and for the open-source implementations on the same CPU instance type and L4 GPU (gray; ISCE3 scored as the mean over its own three patches, times marked est. are estimates, Section 5.1). Torchbp and GRDL, whose backprojections produced no usable image, are drawn off scale. The line is the Pareto front of the FastSAR configurations on these pooled region errors (the bold rows of Table 4 use whole-image errors); those off the front are drawn faded, and Table 4 identifies each configuration. Each dotted line marks the worst FastSAR configuration of one of the image-quality regimes 1 to 3 of Section 5.2.

Of the four devices, the L4 forms a float32-class image (Section 4) at the lowest cost, \$0.75 per thousand Panama images (Figure 1, Table 4). Reduced precision (float16 throughout on the L4, single-pass products on the TPUs) lowers the cost and raises the error (Sections 5.2 and 5.3). On five Capella collections in three modes the float32-class errors stay below -59 dB (Table 5).

2 Related work

Backprojection on parallel hardware has been studied since at least 2009, mostly on GPUs but also on FPGAs [3] and many-core CPUs [4]. Fasih and Hartley gave an early single-GPU CUDA implementation [5], and Benson, Campbell and Cook scaled it to gigapixel images on GPU clusters [6]. Capozzoli, Curcio and Liseno replaced the interpolation of GPU backprojection by a NUFFT [7], and Xu and colleagues ran it in float16 [8].

Fast factorized backprojection descends from McCorkle’s quadtree [9] and Ulander’s formulation [10], and has been applied to nonlinear tracks [11] and to circular SAR on GPUs [12]. Polar format is the standard spotlight image formation algorithm in the textbooks [13, 1]; Jakowatz and

Doren compared it with backprojection [14], and its scene-size limit was derived by Doerry [15] and by Gorham and Rigling [16].

Open-source implementations exist for the CPU (RITSAR [17], GRDL [18], ISCE3 [19] and the AFRL MATLAB toolbox of Gorham and Moore [2]) and for the GPU (torchbp [20] and ISCE3’s CUDA version). The AFRL toolbox, distributed by the NGA, contains the backprojection `bpBasic` and the polar format `pfa_mem`. Outside machine learning the TPU [21, 22] has been used where a problem reduces to dense products, for Fourier transforms [23, 24] and linear algebra [25]. No published work was found that forms SAR images on a TPU or compares implementations on matched hardware against a common float64 reference.

3 FastSAR

FastSAR takes a phase history, the antenna positions and an output grid and returns a complex image; `form_image` forms one image, the `ImageFormer` and `ExactFormer` classes plan once for repeated images of one geometry, and `read_cphd` builds the arguments from a CPHD file. The CPU, CUDA and TPU backends share the host-side float64 geometry and the factorization plan, including its decimation filters. Stripmap and sliding-spotlight collections are formed as mosaics of factorized patches.

Factorized backprojection (factorized BP in the tables) divides the image into tiles, re-centers the phase history on each tile with a phase ramp, filters and decimates it, and recurses [9, 10]. Exact backprojection (exact BP) sums over pulses the range-compressed pulse at each pixel’s range [2], by default with cubic interpolation of profiles oversampled 4 times. Polar format resamples the data onto a rectangular wavenumber grid and forms the image by a two-dimensional FFT [13, 1], with a final resampling that removes its planar-wavefront displacement (Appendix E).

The factorized kernels are written in C++ with OpenMP for the CPU, in CUDA for Nvidia GPUs and in Pallas, JAX’s kernel language, for TPUs (Appendix D). In the profiled development builds, each stage runs within 1.0 to 2.5 times a lower bound set by the hardware unit that limits it, as measured by microbenchmarks on the same device (Table 8).

A pure JAX implementation of the factorized algorithm (the JAX program) preceded the kernels and remains in the library; in the release records it is timed on the CPU only. The precision options are float32 or float16 throughout (float16 data and tensor-core operands, float32 accumulation) on the GPU and three-pass or single-pass bfloat16 products on the TPU (Appendix B). Exact backprojection has no TPU kernel; on a TPU it runs as a JAX program, not timed here, whose scattered reads would take an estimated 4.5 h for the Panama image (Appendix C).

4 Data and protocol

Three spotlight collections from Umbra’s open data catalog [26] are used throughout (Table 1, Figure 2): the Pacific entrance of the Panama Canal (2023-07-18), central Melbourne (2023-02-08) and farmland near Johnston, Iowa (2023-10-20). Each was processed from the CPHD as delivered, on the vendor’s image grid. The four Panama regions of Figure 3 were identified against Sentinel-2 imagery (Appendix I); the lock, port and ship regions are used for the comparison with open-source implementations. Five Capella collections [27] extend the evaluation to a second vendor and to other modes (Section 5.4): two spotlights (2024-10-04 and 2025-03-03), a sliding spotlight (2022-09-30) and two stripmaps (2021-11-12 and 2025-12-08). The sliding spotlight is Capella’s mode SS, labeled dynamic stripmap in its CPHD file.

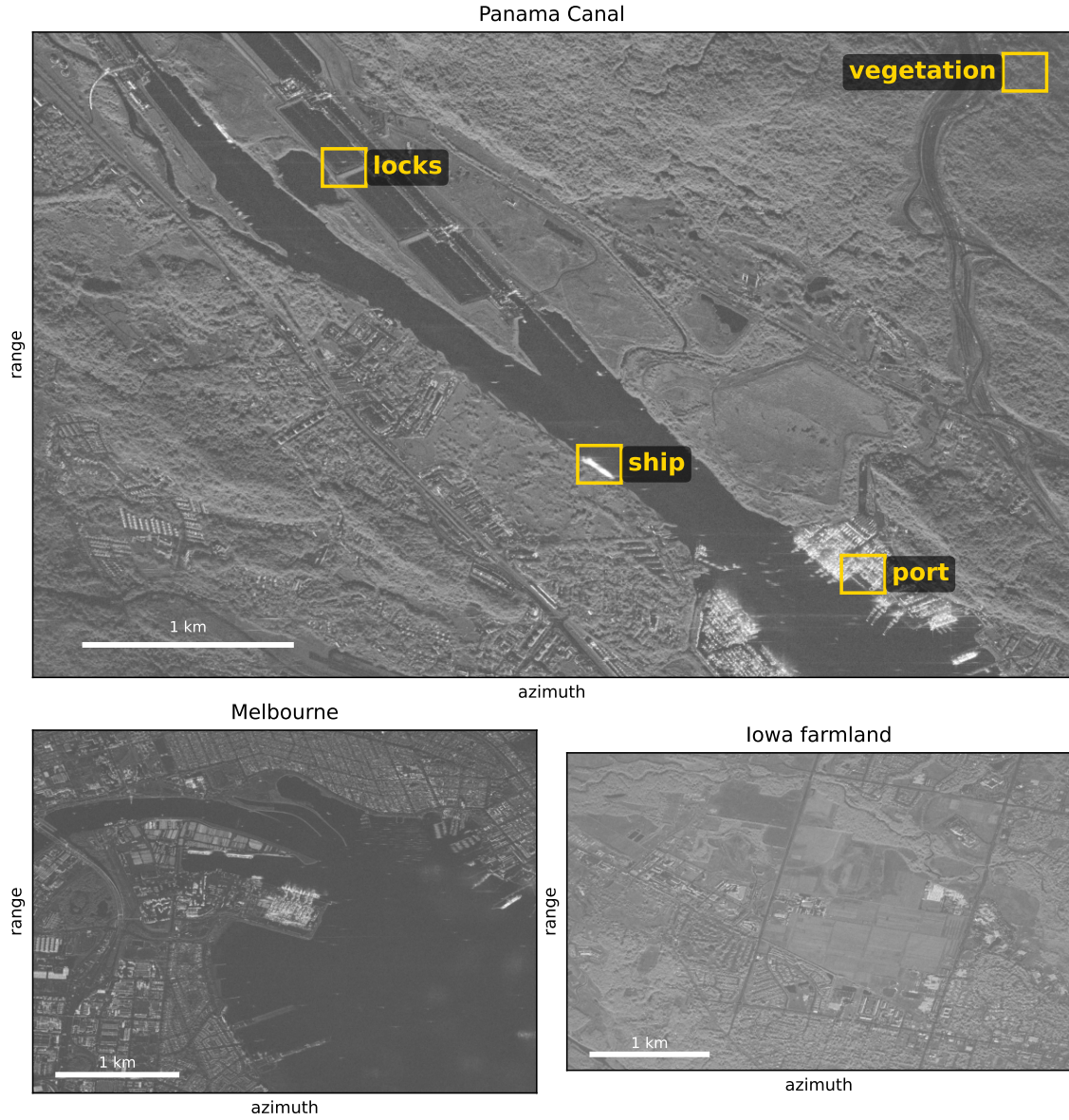


Figure 2: The three collections as formed by the float64 reference, downsampled for display (amplitude in decibels over a 45 dB range, azimuth horizontal, range vertical, 1 km bar). The yellow boxes on the Panama Canal image are the four 512 by 512 pixel regions of Figure 3.

Table 1: The three collections and their native image grids. Spacings are azimuth by range in the slant plane; squint is the Doppler cone angle less 90° at the aperture center; range is the slant range to the scene center at the start of the aperture.

Scene	Image (az $\times$ rg)	Spacing (m)	Pulses $\times$ samples	Bandwidth (MHz)	Grazing angle ($^\circ$)	Squint ($^\circ$)	Range (km)
Panama Canal	$12,207 \times 8,808$	0.413×0.356	$15,186 \times 14,399$	337	42.6	-28.1	738
Melbourne	$10,077 \times 10,273$	0.422×0.298	$15,948 \times 14,399$	405	43.4	0.6	760
Iowa farmland	$8,521 \times 8,465$	0.517×0.328	$13,003 \times 11,663$	356	50.3	2.8	653

Table 2: Instances, the arithmetic timed on each (Appendix B) and on-demand prices (US dollars per hour, us-central1).

Device	Instance	Arithmetic timed	Price (\$/h)
TPU v5e	1 chip (v5litepod-1), 24 vCPU host	bfloat16 single- and three-pass products; float32 data	1.200
TPU v6e	1 chip (v6e-1), 44 vCPU host	bfloat16 single- and three-pass products; float32 data	2.700
L4	g2-standard-4, 1 Nvidia L4 (24 GB)	float32; float16 throughout	0.707
CPU	c4d-highmem-16, AMD EPYC 9B45, 8 cores, 126 GB	float32	0.964

All CPU times, for FastSAR and for the open-source implementations, were measured on c4d-highmem-16 instances with 8 cores and 16 vCPUs (the CPU instance). All GPU times are for an Nvidia L4 (Table 2), hosted by a g2-standard-4 instance for FastSAR on Umbra, a g2-standard-8 for the open-source codes and a g2-standard-32 for the Capella collections (Appendix H).

Images are compared, after one fitted gain, with the reference, a float64 exact backprojection of the same collection with range profiles oversampled 16 times. Once the vendor’s image is resampled through the planar-wavefront displacement model, the reference agrees with it in position to within 0.25 pixel RMS (Appendix F). The error is the energy of the difference relative to that of the reference, in dB, over the whole image unless stated otherwise; the coherence and the amplitude and phase deviations are tabulated with it (Appendix J).

All FastSAR times and images come from calls of the library’s public interface (`ImageFormer`, `ExactFormer` or `form_image`) in release 0.1.0, or in a release candidate whose library code differs from it only as stated in Appendix H.

Time is the wall-clock time to form one image on a warm device, from the phase history in host memory to the image in host memory. It includes transfers to and from an accelerator and the per-collection host work, and excludes compilation and reading from disk. The stage times of Table 8, the build-to-build changes of Table 9 and the polar-format variants of Table 10 are development measurements, as are others marked where they appear; the last row of each device in Table 9 is a release time. Errors below about -69.2 dB are not resolved by the reference (Appendix H).

Cost per image is the hourly price divided by the throughput of a loop that forms images back to back or, where no loop was run, the price multiplied by the time. The images are formed back to back on the CPU and the L4. On the TPUs the next phase history is staged on a worker thread while the current image forms.

Except for the linear-interpolation variants of exact backprojection, each configuration in the tables is the library’s default for its backend and precision option, the fastest of the implementation variants whose errors agreed within 0.1 dB in development. Float32-class accuracy means float32 arithmetic on the CPU and L4 and three-pass products on the TPUs. The factorized kernels were tuned against the bounds of Table 8; polar format was not tuned, and its costs are upper bounds (Appendix H).

Table 3: Error against the reference and time per full image on the Umbra Panama collection. Errors (dB, Section 4) are the range over the lock, port and ship regions of Figure 3, worst to best; ISCE3 was scored instead on three 256 by 256 pixel patches of its own grid (the scene center and two off-center positions), after removal of a linear phase ramp fitted to each. An error of 0.0 dB means that the difference has the same energy as the reference. $16\times$, $4\times$ and $2\times$ give the oversampling of the range profiles; FastSAR’s exact backprojection interpolates them cubically, the others linearly. Times are measured, except those marked est., which scale the time on the scored regions by the number of pixels; see Section 5.1 for the same scaling applied to FastSAR’s exact backprojection.

Implementation	Instance	Error (dB)	Time per full image
RITSAR, as published, $16\times$	CPU	−25.3 to −27.3	44 h (est.)
bpBasic , FFT 2^{17} (default)	CPU	−38.1 to −43.2	57 h (est.)
bpBasic , FFT 2^{18}	CPU	−52.9 to −55.4	69 h (est.)
ISCE3 backproject	CPU	−31.5 to −54.9	4.8 h (est.)
GRDL, factorized	CPU	no image [†]	77.5 s
FastSAR, exact, $4\times$	CPU	−59.9 to −73.8	403 s
FastSAR, factorized, float32	CPU	−56.8 to −61.2	11.3 s
ISCE3 backproject , CUDA	L4	−31.5 to −52.2	27 min (est.)
torchbp, exact, $2\times$	L4	no image* (0.0)	16.0 s
torchbp, factorized	L4	no image*	out of memory
FastSAR, exact, $4\times$	L4	−59.9 to −73.8	19 s
FastSAR, factorized, float32	L4	−56.7 to −61.1	3.8 s
FastSAR, factorized, float16 throughout	L4	−55.3 to −60.4	3.5 s

*torchbp evaluates the pixel-to-antenna distance in float32, whose spacing at the 744 km mid-aperture range of this collection is 6 cm, about two wavelengths, so the phase of each pulse is lost; a float64 evaluation of the same formula forms the image. On the three regions alone, where memory suffices, its factorized backprojection returns an empty image for the same reason.

[†]GRDL’s factorized backprojection is a stripmap algorithm; its image of this spotlight collection is smeared in cross-range (Appendix G).

5 Results

5.1 Comparison with open-source implementations

ISCE3 is the fastest of the open-source backprojections that produced an image. It would need an estimated 4.8 h for the Panama image on the CPU and 27 min on the L4, compared with 11.3 s and 3.8 s for FastSAR’s factorized backprojection (Table 3). After removal of a fitted linear phase ramp, ISCE3 reaches −54.9 dB on the CPU at the scene center and −31.5 to −31.9 dB off center (Appendix G). Of the open-source codes scored on the same three regions, the lowest error is that of **bpBasic** with a 2^{18} -point FFT, 3.9, 4.7 and 6.7 dB above FastSAR’s float32 factorized image on the CPU in the lock, port and ship regions.

The two open-source polar-format codes, **pfa_mem** (44 s) and GRDL’s (24.5 s), leave the planar-wavefront displacement uncorrected. Their log-amplitude correlations with the reference, 0.80 to 0.89 and 0.78 to 0.93, are below the 0.996 to 0.999 of FastSAR’s polar format (Table 12).

FastSAR’s own exact backprojection forms the full image in 403 s on the CPU and 19 s on the L4, 43 and 86 times faster than the ISCE3 estimates, with region errors of −59.9 to −73.8 dB. The same pixel-count scaling, applied to FastSAR’s exact backprojection on the CPU, predicts 17 min, 2.6 times the measured time; the open-source estimates, and the ratios computed from them, are treated as upper bounds.

5.2 Image quality

Each image is also assigned to one of three regimes by its amplitude deviation and its coherence with the reference over 5 by 5 pixel windows. These local statistics respond to isolated errors that the energy ratio averages out. Regime 1 requires an amplitude deviation below 0.05 dB at the 99th percentile and a coherence above 0.999 at the 0.1 percentile. In regime 2 the coherence falls below 0.999 only beside bright returns, and in regime 3 it falls below 0.99 in stripes (Table 14).

On Panama only single-pass TPU products fall in regime 2 and only polar format in regime 3; every other configuration meets the regime 1 criteria (Figures 3 and 4, Table 14). The float32-class images measure -59.9 dB on the L4, -60.0 dB on the CPU, and -57.8 and -57.9 dB with three-pass products on the v6e and v5e. The JAX program on the CPU, which like the TPU kernels evaluates the last level from float32 geometry, gives -58.3 dB, so that geometry may account for most of the TPU images’ deficit of up to 2.2 dB.

Relative to the float32-class images, float16 throughout on the L4 adds 1.0 dB of error and single-pass products on the TPUs add 12 dB. The phase deviation of the single-pass v6e image is 1.1° at the 99th percentile but reaches 106° beside bright returns (173° on Melbourne); its effect on interferometric products was not evaluated. Polar format measures -32.1 dB, its error growing from -40 dB in the inner ring to -30 dB in the outer ring (Appendix E).

FastSAR’s exact backprojection measures -70.0 dB on both the L4 and the CPU, at the resolution limit of the reference. Its margin over the float32 factorized images, 10.0 to 13.1 dB on the three collections (Melbourne and Iowa in the supplementary tables), is therefore a lower bound. With linear interpolation, the scheme of the open-source codes, and profiles oversampled 8 times, it measures -56.9 dB.

5.3 Cost per image on CPU, GPU and TPU

At float32-class accuracy the cost of a thousand Panama images (Table 4) rises from the L4 (\$0.75) through the v5e (\$1.32) and the v6e (\$1.58) to the CPU (\$3.04). The order is the same on Melbourne and Iowa, where the L4 costs \$0.78 and \$0.50 and the CPU \$3.38 and \$2.02 (Table 15). The v6e is the fastest device at this accuracy, 3.1 s per image compared with 3.8 s on the L4, at 3.8 times the L4’s hourly price.

Relative to the float32-class configuration on the same device, float16 throughout costs 9% less on the L4, and single-pass products cost 25 and 38% less on the v6e and v5e. Polar format on the L4 costs \$0.61 per thousand images. On the TPUs it takes 20 to 35 times as long as the three-pass factorized image; its resampling steps are scattered gathers, slow on the TPU’s vector unit (Appendix C). Exact backprojection on the L4 costs \$3.75 per thousand images, 5.0 times the float32 factorized cost.

5.4 Other collections and modes

On the Umbra collections of Melbourne and Iowa, the errors of the factorized configurations are within 1.1 dB of the Panama values (supplementary tables). On the five Capella collections (Table 5), FastSAR’s worst-region error over the three devices lies between -59.1 and -61.5 dB. The sliding spotlight and the stripmaps are mosaics of factorized patches, each formed with the pulses whose beam covers it.

On the CPU instance, ISCE3’s backprojection would take an estimated 36 to 3,600 times as long as FastSAR. ISCE3 places the receiver at the transmit position advanced by the velocity times the delay, whereas the CPHD files give the transmit and receive positions of every pulse, whose midpoint FastSAR and the reference use (Appendix A). Against the reference formed with the

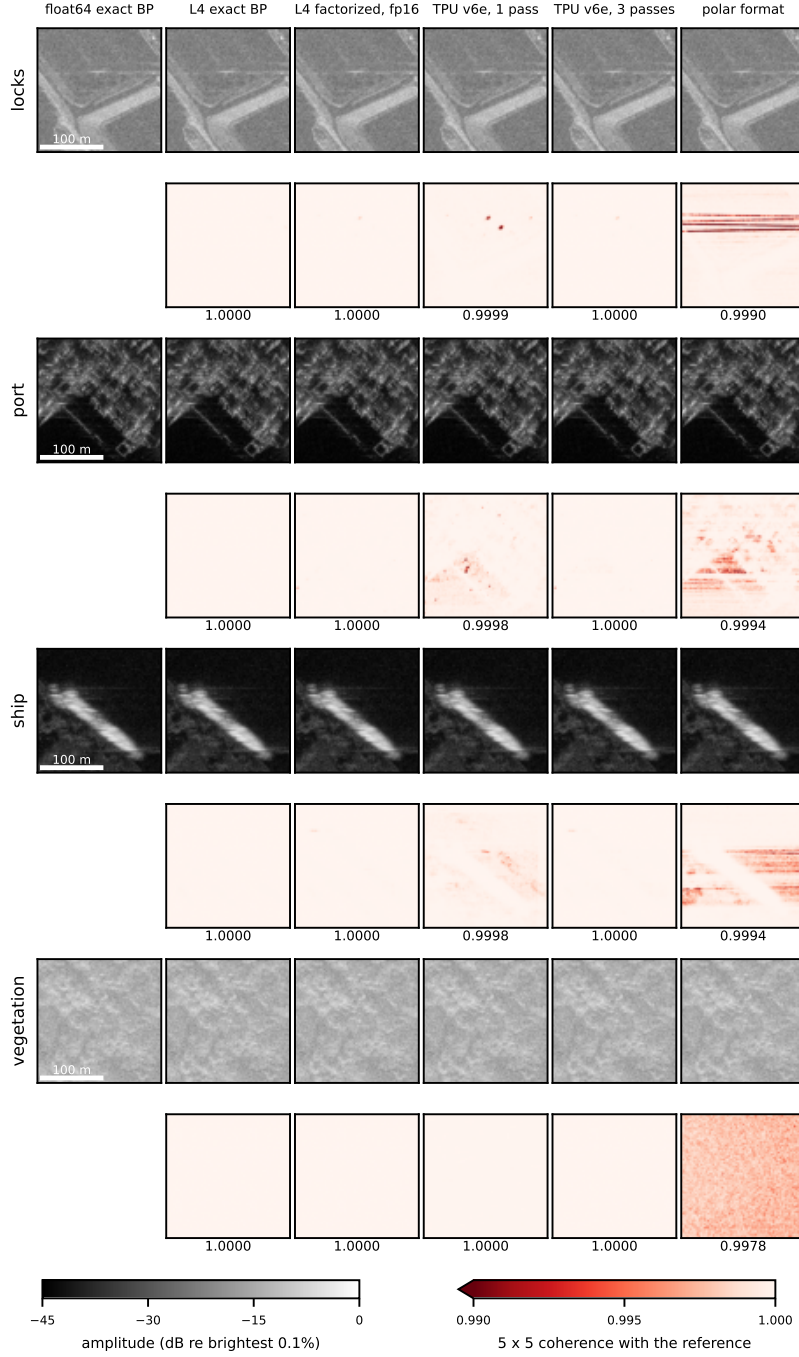


Figure 3: Panama Canal, four regions of 512 by 512 pixels at full resolution (100 m bar). Top of each pair: amplitude over a 45 dB range for the float64 reference, the L4 exact backprojection, the L4 factorized image, float16 throughout, the TPU v6e factorized image with single-pass and three-pass products, and polar format on the L4. Bottom: the 5 by 5 coherence of each image with the reference from 0.99 (red) to 1 (white), with its mean below each panel. The regions are the Cocolí Locks, the Port of Balboa, a ship in the channel and vegetated terrain 2.7 km from the scene center; Appendix I locates them.

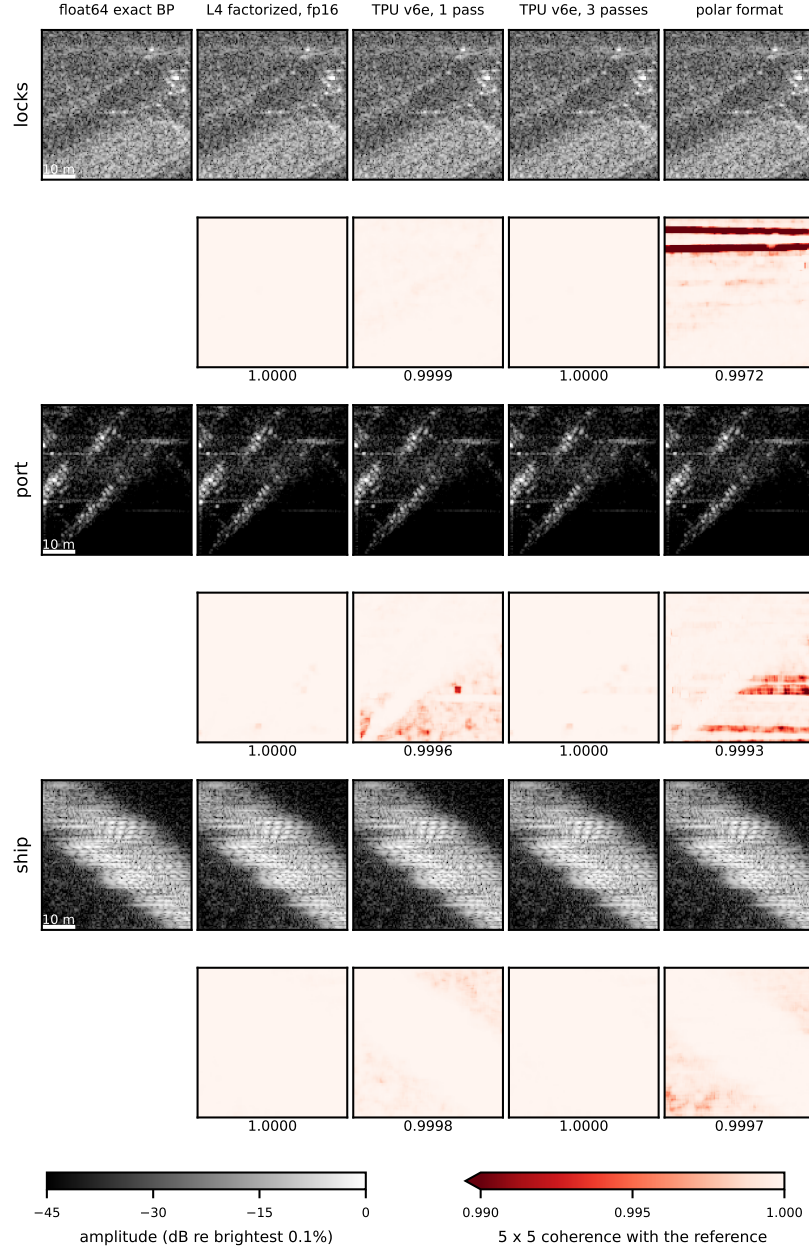


Figure 4: Panama Canal at pixel scale: 128 by 128 pixel windows (53 m in azimuth by 46 m in slant range, 10 m bar) around the brightest feature of the lock, port and ship regions of Figure 3, drawn without interpolation so that each pixel (0.41 by 0.36 m) is visible on the page. Columns, scales and coherence maps as in Figure 3, without the L4 exact backprojection.

Table 4: Panama Canal: the configurations of release 0.1.0, cheapest first. Bold rows are the Pareto front of the whole-image errors, the configurations that no other configuration beats on both cost and error (errors within 0.05 dB before rounding counted as equal). Time is the single-image time; cost is computed from the throughput of the timed loop, which on the TPUs overlaps staging with formation; error is the whole-image error (Section 4); an asterisk marks rows timed without a loop, with cost computed from the single-image time. Energy is that of the processor alone over the per-image time of the cost column; its basis on each device is given with Table 13. Table 15 gives the other two collections.

Device	Algorithm	Arithmetic	Time (s)	\$ per 1000	kWh per 1000	Error (dB)
L4	Polar format	float32, gather	3.0	0.61	0.04	−32.1
L4	Factorized BP	float16 throughout, CUDA kernels	3.5	0.68	0.06	−58.9
L4	Factorized BP	float32, CUDA kernels	3.8	0.75	0.07	−59.9
TPU v5e	Factorized BP	single-pass products, fused kernels	3.6	0.82	0.08 to 0.14	−46.2
TPU v6e	Factorized BP	single-pass products, fused kernels	2.6	1.18	0.09 to 0.15	−46.2
TPU v5e	Factorized BP	three-pass products, fused kernels	5.2	1.32	0.13 to 0.22	−57.9
TPU v6e	Factorized BP	three-pass products, fused kernels	3.1	1.58	0.12 to 0.21	−57.8
CPU	Factorized BP	float32, C++ kernels	11.3	3.04	0.08	−60.0
CPU	Polar format	float32, gather	11.4	3.06	0.08	−32.1
L4	Exact BP	cubic interpolation, 4× oversampled	18.7	3.75	0.38	−70.0
L4	Exact BP	linear interpolation, 8× oversampled	20.3	3.93	0.40	−56.9
TPU v5e	Polar format	float32, gather	103.4	34.45	3.44 to 5.74	−32.1
TPU v6e	Polar format	float32, gather	110.2	82.69	6.12 to 10.72	−32.1
CPU	Exact BP	cubic interpolation, 4× oversampled	403.0	107.90*	2.80	−70.0

CPHD positions, ISCE3’s worst region is -25.8 to -50.6 dB, 10 to 34 dB above FastSAR’s. Scored against a reference formed with its own receive model, ISCE3’s worst region is 4.0 to 16.6 dB below FastSAR’s worst error over the three devices on the 2024 spotlight, the 2025 spotlight and the 2021 stripmap.

GRDL’s frequency-domain processors, whose slant-plane images are not scored against the reference, are faster than FastSAR on the CPU for both stripmaps (range-Doppler, 27.7 s compared with 47.5 s on the 2021 collection) and slower for both spotlights. The v6e has the shortest time on all five collections. The L4 costs less per image than the v6e on all collections except the 2025 spotlight; a thousand images of the 2025 stripmap cost \$95 on the L4, \$130 on the v6e and \$234 on the CPU.

The 2021 stripmap and the 2024 spotlight were also compared with the vendor’s image on the vendor’s pixel grid, by the method of Appendix F. The correlation of linear amplitudes is 0.993 for the stripmap mosaic (165 patches, 512 by 512 pixel crop) and 0.78 for the spotlight formed by exact backprojection (128 by 128 pixels at the scene center). Neither is comparable with the windowed log-amplitude correlation of Table 11. In both of these CPHD files the phase sign declared in the metadata is opposite to the sign with which the data focus, and FastSAR’s reader reverses it.

Table 5: The five Capella collections. FastSAR time per image (host memory to host memory, Section 4) at float32-class accuracy and cost per thousand images (hourly price times the single-image time) on the CPU instance, the L4 and the v6e; the L4 runs used a g2-standard-32 instance at \$1.734 per hour. * Run with smaller first-level groups (with a `MemoryWarning`), because full speed needs more than the roughly 23 GB free on the L4: 33.8 GB for the 2024 spotlight, 23.3 GB for the 2025 one. FastSAR error: the worst of three 512 by 512 pixel regions against the float64 reference over the three devices. ISCE3: whole-image time estimated from the three regions on the CPU instance, and worst region error against a reference formed with ISCE3’s receive model (Section 5.4) and against the reference with the per-pulse antenna positions of the CPHD file (CPHD), as for FastSAR. GRDL: algorithm (PFA, polar format; RDA, range-Doppler; SMPFA, GRDL’s stripmap polar format) and time on the CPU instance; its slant-plane images are not scored against the reference.

Collection	Pulses	Mpx	FastSAR time (s)			FastSAR cost (\$/1000)			FastSAR error (dB)	time (h)	ISCE3			GRDL	
			CPU	L4	v6e	CPU	L4	v6e			own	model (dB)	CPHD (dB)	alg.	time (s)
Spotlight, 2024	74,203	258	54.0	18.0*	12.3	14	8.7*	9.2	-59.1	54.1	-63.1	-25.8	PFA	99	
Spotlight, 2025	54,267	255	46.1	17.9*	9.8	12	8.6*	7.4	-59.2	41.6	-75.7	-36.3	PFA	70	
Sliding spotlight, 2022	123,799	72	286	57.7	55.3	77	28	41	-61.5	2.9	-61.3	-38.6	-	-	
Stripmap, 2021	39,898	83	47.5	11.5	9.6	13	5.5	7.2	-60.2	2.3	-64.8	-50.6	RDA	28	
Stripmap, 2025	237,720	1132	875	197	173	234	95	130	-60.6	48.2	-59.2	-26.9	SMPFA	550	

6 Conclusions

On the CPU and the L4, where both algorithms were timed, factorized backprojection formed the full spotlight images 4.6 to 4.9 times faster than exact backprojection on the L4 and 36 times faster on the CPU on Panama. Its whole-image error was at least 10.0 dB higher. FastSAR’s exact backprojection is itself 43 times faster than ISCE3’s estimated time on the CPU.

On the Capella collections the comparison with ISCE3 depends on the receive model. FastSAR has the lower error on every collection when both are scored with the CPHD antenna positions. When ISCE3 is scored against a reference formed with its own receive model, its error is lower on three of the five. Torchbp, which evaluates absolute range in float32, did not form an image of the Panama collection.

The four devices form a float32-class Panama image in 3.1 to 11.3 s, and the cost is lowest on the L4. Float16 throughout lowers it by a further 9% and raises the error by 1.0 dB. Single-pass TPU products save 25 to 38% but leave phase errors of up to 173° beside bright returns, and three-pass products are recommended on the TPUs when the image phase is used, as in interferometry. The cost comparison uses on-demand prices only (Appendix H).

Acknowledgments. The code and the text of this paper were drafted with the assistance of a large language model; the authors designed the experiments, ran and checked every measurement, and take full responsibility for the content.

Code and data. FastSAR is available at <https://github.com/saulpingerman/FastSAR> and, as release 0.1.0, from the Python Package Index (`pip install fastsar`). The cloud run scripts, the measurement records behind every table and the source of this paper are at <https://github.com/saulpingerman/sar-accel-study>. The radar data are from the Umbra [26] and Capella [27] open data programs and the optical images from Copernicus Sentinel-2 [28]; none is redistributed, and the scripts fetch them from their public sources.

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A Algorithms

The phase history is motion compensated to the scene reference point. For pulse p with antenna position $\mathbf{a}_p$ and frequency sample f_k ,

$$S_{p,k} = \sum_n \sigma_n \exp\left(-j \frac{4\pi f_k}{c} \Delta R_p(\mathbf{x}_n)\right), \quad \Delta R_p(\mathbf{x}) = \|\mathbf{x} - \mathbf{a}_p\| - \|\mathbf{a}_p\|, \quad (1)$$

where σ_n and $\mathbf{x}_n$ are the complex amplitude and position of scatterer n and c is the speed of light. The CPHD files give the transmit and receive positions per pulse, whose midpoint was used as $\mathbf{a}_p$, and the start frequency and sample spacing per pulse; the mean start frequency and spacing were used, which differ from the per-pulse values by at most 0.02 of a sample spacing. A Taylor weighting ($\bar{n} = 4$, -35 dB sidelobes) [29] was applied along both axes in every algorithm. The image grid is the vendor's: pixel (i, j) lies at $(i - n_x/2) s_x \mathbf{e}_1 + (j - n_y/2) s_y \mathbf{e}_2$ from the reference point, with $\mathbf{e}_1$ and $\mathbf{e}_2$ the SICD column and row unit vectors and s_x, s_y its sample spacings.

Exact backprojection. In the reference each pulse is range compressed by a zero-padded inverse FFT with 16 times oversampling, and each pixel is the sum over pulses of the linearly interpolated profile at ΔR_p times the carrier phase $\exp(j4\pi f_c \Delta R_p/c)$ [2]. The range difference is evaluated in the cancellation-free form

$$\Delta R_p(\mathbf{x}) = \frac{q}{1 + \sqrt{1 + q/r_p}}, \quad q = \frac{\|\mathbf{x}\|^2}{r_p} - 2 \hat{\mathbf{u}}_p \cdot \mathbf{x}, \quad (2)$$

with $r_p = \|\mathbf{a}_p\|$ and $\hat{\mathbf{u}}_p = \mathbf{a}_p/r_p$. The reference implementation evaluates everything in float64 with Numba [30] on the CPU. FastSAR's exact backprojection (**ExactFormer**, with C++ and CUDA kernels) evaluates the range, the unit vector and the reciprocal range of each tile center in float64 once per pulse, and each pixel's range as the center's plus a third-order expansion in its offset $\mathbf{d}$ from the center, $\mathbf{d} \cdot \hat{\mathbf{u}} + D/(2r) - (\mathbf{d} \cdot \hat{\mathbf{u}})D/(2r^2)$ with $D = \|\mathbf{d}\|^2 - (\mathbf{d} \cdot \hat{\mathbf{u}})^2$, in float32. The tile, 32 by 32 pixels at orbital range, is reduced where the predicted error of the next term would exceed 3×10^{-4} rad. Each pulse is range compressed by a zero-padded FFT, only the bins its tiles can reach are kept, and the profiles are read with cubic Lagrange interpolation (oversampling 4, the default) or linear interpolation (oversampling 8, or 16 on request). On the L4 the Panama image, 1.6 trillion interpolated reads, takes 18.7 s, or 87 billion reads per second.

Factorized backprojection. The image is divided recursively into tiles, as in the quadtree [9, 31] and fast factorized [10] backprojection algorithms. At each level the phase history of a parent tile is multiplied by $\exp\{j4\pi f_k(\|\mathbf{c} - \mathbf{a}_p\| - \|\mathbf{c}' - \mathbf{a}_p\|)/c\}$ to move its reference point from the parent center $\mathbf{c}'$ to a child center $\mathbf{c}$; the child's signal is then band limited in frequency and in pulse index in proportion to the child's size, and both axes are low-pass filtered and decimated by a Kaiser-windowed sinc [32] with 70 dB design attenuation. At the last level the range to each pixel is linearized about the tile center, the kernel separates in the two image coordinates, and the tile image is a product of two matrices followed by a per-pixel quadratic phase correction, so that no stage requires an indexed read. Rectangular images of any size are supported: the side of each level divides the padded image by factors chosen from 8, 7, 6, 5, 4, 3 and 2, and the decimation factors follow from the range and Doppler extents of a child tile. For the Panama image the three levels divide the azimuth side by 8, 8 and 6 and the range side by 8, 7 and 5, and the first level decimates the 14,399 samples by 6 and the 15,186 pulses by 4; the final tiles are 32 pixels on a side on all three collections, the default size, which the predicted error of each collection geometry supports. In the

Table 6: Floating-point type of each step in the timed configurations. Geometry means tile centers, ranges and phase-ramp coefficients; ramps means the sines and cosines themselves; products means the operands of the matrix products, filters or sums, always accumulated in float32 or better.

Device	Algorithm, label	Data in memory	Geometry	Ramps	Products
L4, CPU	Exact BP, cubic or linear interpolation	float32 profiles	float64 per tile, float32 within the tile	float32	float32 sums
L4	Factorized BP, float32, CUDA kernels	float32	float64	float32	float32 fused multiply-adds (levels and final)
L4	Factorized BP, float16 throughout, CUDA kernels	float16	float64	float32	float16 data into float32 fused multiply-adds in the levels; float16 operands on the tensor cores in the final stage
TPU	Factorized BP, three-pass products, fused kernels	float32	float64 (2 levels), float32	float32	3 bfloat16 passes (levels and final)
TPU	Factorized BP, single-pass products, fused kernels	float32	float64 (2 levels), float32	float32	bfloat16 operands, 1 pass (levels and final)
CPU	Factorized BP, float32, C++ kernels	float32	float64	float32	float32 fused multiply-adds (levels and final)
CPU	Factorized BP, float32, JAX program	float32	float64 (2 levels), float32	float32	float32
all	Polar format, float32	float32	float64	float32	float32 sums over gathered samples; float32 FFTs

Table 7: Floating-point formats. Significant bits include the implicit bit; the spacing is the distance between adjacent representable values near 1.

Format	Bits	Significant bits	Exponent bits	Spacing near 1
float64	64	53	11	2.2×10^{-16}
float32	32	24	8	1.2×10^{-7}
float16 (IEEE binary16)	16	11	5	9.8×10^{-4}
bfloat16	16	8	8	7.8×10^{-3}

JAX program and the TPU kernels the first two levels compute their phase ramps from float64 geometry on the host and the remaining level from float32 geometry on the device (the CUDA and C++ kernels use float64 geometry at every level); during development a float32 second level was found to add an error that grew with the size of the scene. The ramps themselves are evaluated on the device in float32. The decimation filters are applied as dense matrix products on the TPU, as strided convolutions in the JAX program on the CPU, and as explicit fused multiply-adds in the CUDA and C++ kernels. Polar format is described in Appendix E.

B Floating-point formats and the arithmetic of each configuration

Table 6 states, for every configuration that was timed, in which floating-point type each step runs. In development a configuration in which the phase ramps were also computed in bfloat16 produced an unusable image; release 0.1.0 does not offer it.

Table 7 lists the formats that appear in this paper. Float16 (IEEE binary16) has 11 significant bits and a 5-bit exponent. Its relative rounding error is at most 1 part in 2048, and it overflows above 65,504, so the phase history is scaled to unit peak before conversion. Bfloat16 [33] keeps the 8-bit exponent of float32 and only 8 significant bits, 1 part in 256. On the TPU every matrix product is formed from bfloat16 operands. A float32 operand is either rounded once (one pass), or split into a bfloat16 head and tail with the cross terms accumulated (three passes), or split into three parts (six passes, used by the JAX program), which recovers float32 accuracy [34]; the three-pass product keeps the first 16 significant bits of each operand. The L4’s tensor cores accept float16 operands, which the final stage of float16 throughout uses. The CPU kernels use no reduced-precision arithmetic; the AVX-512 bfloat16 instructions of the processor were not evaluated.

The measured image errors agree with the operand resolutions of Table 7. Rounding each operand of the factorized products to bfloat16 adds noise about 48 dB below the operand [35], and the measured error of the single-pass TPU image, once the error of the three-pass image is removed, matches the rounding noise of the two operands accumulated through three levels. The L4’s float16 operands have 18 dB finer resolution than bfloat16, and their rounding adds 1.0 dB to the error of

the float32 CUDA build, whose products are all float32.

C Exact backprojection on the TPU

FastSAR has no TPU kernel for exact backprojection; on a TPU `ExactFormer` runs a JAX program, because the TPU has no fast path for the per-pixel lookup of exact backprojection. In development microbenchmarks, data-dependent one-dimensional gathers ran on the TPU at about 10^8 reads per second, and the scattered two-dimensional gathers of the polar-format resampling at about 4×10^7 (Appendix E); at the first rate the 1.6 trillion reads of the Panama image would take about 4.5 h. A TPU core has a matrix unit, which multiplies dense bfloat16 tiles at the chip’s rated rate, and a vector unit, whose element-wise and gather throughput was roughly two orders of magnitude lower per element in these microbenchmarks; Google’s Cloud TPU documentation (accessed October 2026) describes no texture or cache path for scattered reads. The sparse cores of the v6e, intended for embedding lookups, are documented near 10^9 rows per second per chip, which would still need about 30 minutes for the Panama image. A one-hot matrix product would move the lookup onto the matrix unit, but its matrix has one nonzero in several thousand entries, and the dense product would exceed the work of the factorized algorithm by a comparable factor. The closest published work computes FFTs and nonuniform FFTs, the latter for MRI reconstruction, as matrix products on the TPU [23, 24].

D Kernels for the factorized algorithm

The factorized rows of Table 4 are formed by kernels written for each device after a stage-by-stage profile of the XLA code generated from the JAX program; the time each development build removed is in Table 9 and the bounds the profiled builds reach in Table 8. In development, on the device, the JAX program took 4.6, 9.3 and 13.7 s for the float32-class Panama image on the v6e, v5e and L4, and the last development builds 2.4, 4.6 and 3.5 s, factors of 1.9, 2.0 and 3.9. On the CPU release 0.1.0 forms the image in 224 s with the JAX program and 11.3 s with the C++ kernels, a factor of 20. The float32 images of the CUDA and C++ kernels are also closer to the reference than that of the JAX program, a difference consistent with their float64 geometry at every level. In the terms of Appendix A, a level splits each parent tile’s phase history into those of its children, the tiles of the next level. Level 0 splits the whole collection into the first-level tiles; levels 1 and 2, which run inside one first-level tile per device call, are called the tile levels below. In the JAX program and the TPU kernels only level 2 takes its geometry from the device; the CUDA and C++ kernels compute it in float64 on the host. The plan and filters are the same in every build. Each build’s image was checked against the dense image at the same precision on a simulated scene before it was timed.

Diagnosis. In development, on the v6e, one first-level tile of the Panama image took 51 ms in the JAX program with single-pass products: 8.3 ms in level 0, 13.3 in level 1, 7.9 in level 2 and 21.4 in the final stage. The three levels are bound by memory traffic: XLA writes the rotated phase history of every child (24 bytes per sample, read back by the decimation product) at about 580 GB/s, a third of the v6e’s nominal 1,640 GB/s (1,391 GB/s measured, below), and the children of the three levels total 63 billion samples per image. The final stage materializes the two ramp matrices of every 32 by 32 tile, $2T$ by $P_f Q_f$ each, through five or six element-wise passes before the one product that uses them, and its direct form evaluates 400,000 sines and cosines per tile. The same structure holds on the L4 at the convolution path’s lower bandwidth.

TPU kernels (Pallas). Two Pallas kernels replace the XLA stages. The level kernel takes a block of pulses and a window of columns of a parent into the core’s vector memory and applies the phase ramps of up to eight children in turn. Each ramp is the product of a coarse table, one angle per 128-column chunk, and a fine table, one angle per column within the chunk; both are precomputed once per pulse and child by XLA, so the kernel evaluates no sines (Pallas lists them as its most expensive operations, and an earlier build that evaluated the coarse table in the kernel spent a third of its time on them). The rotated window is multiplied by the banded block of the decimation matrix on the matrix units, in one or three bfloat16 passes, and at the two tile levels the pulse-axis decimation is accumulated on the matrix units across the pulse blocks in the same kernel, so neither the rotated window nor the frequency-decimated child reaches high-bandwidth memory; at level 0 the decimated child is too large for vector memory and the pulse product stays a separate dense product. Children of one parent share each load, the tile levels batch the parents of a tile into one launch, and first-level tiles are formed in groups so that the phase history itself is read once per group (groups of 16 in the development builds of Table 9; release 0.1.0 uses groups of 4 and also applies the level-0 pulse decimation as a banded product). Where one block of outputs covers a whole row (the tile levels), the window is clipped to the data and read once; at level 0 consecutive output blocks overlap by the filter length.

The final kernel forms the $2T \times 2T$ product of Appendix A ($T = 32$ pixels on a side, $P_f \times Q_f$ pulses by frequency samples per tile) on the matrix units, with both ramps generated by a complex recurrence along the frequency index q : for a pixel offset d along an image axis and the component u_p of the unit vector to the antenna along that axis, the ramp $\exp(-j2\pi du_p(a_0 + a_1q))$, with a_0 and a_1 the cycles per meter at the first frequency sample and per sample step, is XW^q with $X = \exp(-j2\pi du_p a_0)$ and $W = \exp(-j2\pi du_p a_1)$, so one tile needs four tables of sines and cosines of size $T \times P_f$ instead of one sine per sample. A first version of this kernel, with one sine per sample, was no faster than the XLA final stage (Table 9), because the sines kept the stage bound by the transcendental unit. The recurrence removes these sines. Building the level kernel’s fine tables in XLA by successive doubling nearly doubled the image time, because each intermediate table of fewer than 128 columns occupies full 128-lane tiles in TPU memory. Evaluating the coarse-table sines over the used lanes only and padding afterwards cost 4 to 8%, since XLA holds the table 128 lanes wide either way and the pad is a copy. Stacking four tiles into one 256 by 256 product in the final stage, to fill the systolic array, gained 3% on the v6e at three-pass precision and nothing otherwise, and lost 2 to 4% on the v5e, whose arrays are 128 by 128.

Bounds. Table 8 compares each stage with the bound set by the unit that binds it, for the profiled builds of Table 9, timed on the device; the released library’s times from host memory to host memory are those of Table 4. The unit rates were measured on each device with microbenchmarks in the same framework (`ubench_tpu.py`, `ubench_gpu.py` in the repository): on the v6e, high-bandwidth memory copies at 1,391 GB/s; the vector unit executes the level kernel’s own rotation mix on data resident in vector memory at 117 billion samples per second and a plain copy through vector memory at 155 billion, so the rotation runs within 1.3 times of the vector load and store path itself; XLA evaluates 57 billion sine and cosine pairs per second; the matrix units reach 457 TFLOP/s on a large bfloat16 product and 42 at the final stage’s 64-row shape. Against these, on the v6e, level 1 runs at the vector unit’s rate (ratio 1.0), level 0 at 1.4 times its vector bound with the unfused pulse product passing through memory, and level 2 at 2.5 times its largest bound but at 0.9 of the sum of the terms that run in sequence: the rotation (0.24 s per image), the sines XLA evaluates for its tables (0.25 s; the level’s rows are 352 samples long, so each child and pulse needs 256 sines for 512 rotated samples) and the traffic of those tables (0.17 s). The final stage is 2.2 times its matrix

bound, the excess lying in the ramp recurrence on the vector unit, which the microbenchmarks did not isolate. On the v5e the matrix units, about three times slower than the v6e’s on a large product against a vector unit only 12% slower, become the binding unit of levels 0 and 1, and every stage sits between its largest bound and the sum of its bounds (1.5 to 2.2 times the former, 0.6 to 0.8 of the latter), so its units overlap only partly. The largest remaining inefficiency identified within the factorization is the level-2 table generation, which angle-addition tables could reduce by about a factor of eight, saving roughly 0.2 s of the 1.7 s device time of the v6e image; every other TPU stage is at most 2.2 times its largest single bound and, where a sum is defined, below the sum of its bounds.

L4 kernels (CUDA). On the L4 the two kernels are written in CUDA and compiled through CuPy [36]. The level kernel loads the window of 32 pulses into shared memory with consecutive threads on consecutive columns, so that global loads are coalesced, and rotates it in place for each child, the rotation of each thread’s samples generated as a geometric sequence. Each thread then computes two adjacent outputs of one pulse with the 44-tap decimation filter as fused multiply-adds; for this phase consecutive threads are consecutive pulses, so no two threads of a group read the same shared-memory bank, and the outputs are staged through shared memory and written row by row. The pulse-axis filter is tiled the same way. The final kernel builds the two ramp tables of a tile in shared memory by the same recurrence and each thread accumulates a 4 by 2 block of pixels; in the float16 tensor-core builds the tables are converted to float16 and the $2T \times 2T$ product runs on the tensor cores with float32 accumulation. A Triton version of the level kernel through Pallas, with the 16-column windows its shared-memory limit allows, gained 1 to 4% with float16 products and was slower in float32 (development device timings, within the run-to-run spread of Appendix H); the library uses the CUDA builds.

The CUDA code went through seven builds, five of them profiled (Table 9; the profiles are in the repository). The profile of the second build placed 1.5 s per image in the level kernel, 1.3 in the final stage and 0.5 in the pulse filter. The pulse-major thread mapping of the third build freed the filter reads of bank conflicts but made the global loads and stores of that kernel scattered, and its time doubled; coalesced loads through shared memory, the geometric-sequence rotation and the staged stores brought it to 1.9, 1.7 and 1.4 s in the fourth, fifth and sixth builds. The float32 final stage was bound by the latency of its table-building recurrence and by one shared-memory word per fused multiply-add; with a 4 by 2 block per thread and interleaved complex tables it fell from 1.3 to 1.0 s. The tensor-core version was 0.43 s faster per image on the device than the float32 one in the sixth build. The seventh build stores the phase history, every intermediate and the final-stage operands in float16 and accumulates in float32 inside the filters, which halves the memory traffic that bounds the level kernel and the pulse filter. It is the float16 configuration of Table 4: release 0.1.0 forms the image with it in 3.5 s compared with 3.8 s for the float32 build, with 1.0 dB more error.

Against the L4’s measured rates (memory copy 246 GB/s; the level kernel’s inner mix on resident shared memory 20.2 billion filter outputs per second; the final kernel’s inner mix 6.1 trillion fused multiply-adds per second, 41% of the chip’s peak; coalesced stores 252 GB/s against 69 strided), the pulse filter runs at its memory bound, the level kernel at 1.1 times the sum of its arithmetic and memory bounds (1.9 times their maximum; the two overlap only partly at two blocks per multiprocessor), and the float32 final stage at 2.2 times the bound of its inner mix, the excess being spent in building the tables. The remaining structural cost of the float32 build is the 172 GB per image that the level kernel writes and the pulse filter reads back; a streaming fusion of the two could remove about half of it, worth about 0.3 s of the 3.6 s that the build spends on the device. The float16 build halves that traffic instead; in release 0.1.0 it forms the image 0.3 s faster than the

float32 build (Table 4), less than halving a purely memory-bound stage would give, because the level kernel’s arithmetic bound is close to its memory bound. Float16 storage does not move that bound, because the L4’s CUDA cores issue float16 and float32 fused multiply-adds at the same rate (128 per clock per multiprocessor at compute capability 8.9 [37]). Further gains from float16 on the L4 would depend on the tensor cores, for which the level filters would have to be rewritten as matrix products, as in the TPU kernel.

CPU kernels (C++). The level filter, the pulse filter and the final stage were written in C++ with OpenMP for the 8-core instance, compiled by GCC 13 for AVX-512 with 512-bit vectors preferred and called from the Python driver through a shared library; the host side (plan, filters, float64 geometry) is the code used for the other devices. The design that came closest to the arithmetic bound assigns the children of a parent to vector lanes: a sample of the parent row is a scalar broadcast, its rotation one complex multiply per 16 children with a per-lane phase step, and the filter accumulates eight outputs per child vector in registers, so no strided read occurs. The pulse filter reads each input row once per chunk of 16 output rows, and the final stage builds its two tables three pulses at a time, by a geometric recurrence over the pulse index and another over the pixel index (four sines per pulse and table), so that they stay in the second-level cache while the product consumes them. Four builds were measured (Table 9). The first, a polyphase filter vectorized along the output columns, took 51.8 s per Panama image, because its stride- D gathers cost about twelve instructions per useful multiply-add. The second build, with children assigned to vector lanes, took 19.8 s. Reuse of the intermediate buffers, which removed one page fault per 4 KB of fresh allocation inside the parallel regions, and register-resident accumulators reduced this to 13.2 s in the third. The fourth build, which builds and consumes the final-stage tables three pulses at a time, took 12.4 s, of which 3.8 s was the final stage. The fourth build is the one released; in development it ran on 8 threads, one per core, and 16 threads gave the same time. Against the instance’s measured rates (memory copy 49 GB/s; the filter’s inner loop 314 billion fused multiply-adds per second on resident data; the final product’s inner loop 1.04 trillion fused multiply-adds per second, near the 8 cores’ float32 peak of two 512-bit fused multiply-adds per core and cycle), the level kernel runs at 1.2 times its arithmetic bound (0.8 of the sum of arithmetic and memory), the pulse filter at 1.05 times its memory bound and the final stage at 1.4 times its arithmetic bound; the excess is attributed to table building (Table 8).

Table 8: Each stage of the profiled development builds (on the TPUs and the CPU the last build of each device in Table 9, on the L4 the sixth, float32, build) against the lower bound set by the unit that binds it, Panama image, single-pass or float32 arithmetic. The bounds divide the inherent work of the stage by rates measured on the same device with microbenchmarks in the same framework (vector unit on the rotation mix of the kernel with data resident on chip, high-bandwidth memory copy, matrix units at the product shapes of the kernel, sines in XLA; on the L4 the inner mixes of the kernels on resident shared memory and the memory copy rate). A ratio near one means the stage runs at the rate of that unit; the last column divides the measured time by the sum of all the bounds of the stage, the time it would take if none of its units overlapped. Stage times are measured on the device, excluding transfers.

Device	Stage	Binding unit	Bound (s)	Measured (s)	Ratio	To the sum
TPU v6e	level 0	vector unit	0.15	0.22	1.4	0.60
	level 1	vector unit	0.30	0.30	1.0	0.52
	level 2	sines (XLA)	0.25	0.61	2.5	0.88
	final	matrix units	0.13	0.29	2.2	–
TPU v5e	level 0	matrix units	0.33	0.50	1.5	0.72
	level 1	matrix units	0.38	0.60	1.6	0.62
	level 2	sines (XLA)	0.40	0.88	2.2	0.78
	final	matrix units	0.25	0.44	1.7	–
L4	levels (rotation and frequency filter)	memory	0.73	1.36	1.9	1.08
	pulse filter	memory	0.41	0.46	1.1	1.11
	final (float32)	arithmetic	0.45	0.98	2.2	2.07
CPU	levels (rotation and frequency filter)	arithmetic	4.19	4.87	1.2	0.79
	pulse filter	memory	2.06	2.17	1.1	0.84
	final (float32)	arithmetic	2.67	3.76	1.4	1.35

Table 9: The builds of the factorized algorithm on Panama, two precisions per accelerator and float32 on the CPU: 16-bit means single-pass products on the TPUs and, on the L4, float16 products (JAX program), the float32 build with the float16 tensor-core final stage (CUDA builds four to six) or float16 storage throughout (build seven); float32-class means three-pass products on the TPUs and float32 on the L4. The first row of each device is the device time of the JAX program per image (s), measured in the development session of the first kernel build (on the CPU, which has no transfers, the JAX program of the released library); the rows between give the change each development build made to the device time of the build before it (s; negative is faster). Each change is relative to the nearest row above that is not marked rejected or instead. The last row is the time of the released library from host memory to host memory (Table 4), which includes the transfers and the per-collection host work and is not directly comparable with the device time of the last build (on the CPU, which has no transfers, the release time is the shorter). On the CPU the changes add up from the first C++ build (51.8 s, Appendix D), the step from the JAX program to it being omitted.

Device	Build	16-bit (s)	float32-class (s)
TPU v6e	XLA: ramps materialized, dense decimation products, XLA final stage	3.82	4.60
	Fused level-0 kernel, one child per load	+0.69	+0.65
	Fused level kernels, children share a load, XLA final stage	-0.41	-0.50
	Same, tile level batched over parents	-1.04	-1.03
	Plus fused final stage, per-sample sines	-0.04	-0.02
	Plus fused final stage, ramps by recurrence	-1.02	-0.95
	Plus clipped windows, no halo at levels 1 and 2, precomputed fine tables	-0.11	-0.13
	Fine tables by doubling instead (rejected)	+2.16	+2.16
	Coarse tables precomputed too, pulse decimation fused into the kernel at levels 1 and 2	-0.21	-0.26
	Four tiles per step in the final stage instead (3% faster on the v6e at three-pass precision only; not kept)	0.00	-0.07
	Coarse-table sines over the used lanes only, padded afterwards (rejected)	+0.10	+0.10
	Level kernels of the row above, one-tile recurrence final stage (last development build)	0.00	0.00
	Release 0.1.0, host memory to host memory	2.58	3.12
TPU v5e	XLA: ramps materialized, dense decimation products, XLA final stage	6.49	9.29
	Fused level-0 kernel, one child per load	+1.23	+0.82
	Fused level kernels, children share a load, XLA final stage	-0.94	-1.55
	Same, tile level batched over parents	-1.75	-1.63
	Plus fused final stage, per-sample sines	+0.10	+0.27
	Plus fused final stage, ramps by recurrence	-1.71	-1.62
	Plus clipped windows, no halo at levels 1 and 2, precomputed fine tables	-0.23	-0.49
	Fine tables by doubling instead (rejected)	+3.68	+3.67
	Coarse tables precomputed too, pulse decimation fused into the kernel at levels 1 and 2	-0.43	-0.52
	Four tiles per step in the final stage instead (3% faster on the v6e at three-pass precision only; not kept)	+0.10	+0.10
	Coarse-table sines over the used lanes only, padded afterwards (rejected)	+0.23	+0.23
	Level kernels of the row above, one-tile recurrence final stage (last development build)	0.00	0.00
	Release 0.1.0, host memory to host memory	3.63	5.20
L4	XLA: ramps materialized, cuDNN convolutions, XLA final stage	11.16	13.69
	Triton level-0 kernel (Pallas), 16-column windows (not kept; the CUDA rows below are measured from it)	-0.43	+5.03
	CUDA: shared-memory FIR levels, one pixel per thread final	-	-13.10
	Final stage 2 by 2 pixels per thread	-	-1.01
	Pulse-major filter threads (conflict-free reads, scattered loads and stores), tiled pulse filter, 4 by 2 final	-	+0.84
	Coalesced window loads, children as in-place phase steps; float16 tensor-core final	-6.98	-1.05
	Rotation as a geometric sequence per thread; interleaved complex tables in the final	-0.20	-0.42
	Output tiles staged in shared memory for coalesced stores (last float32 development build)	-0.39	-0.39
	Float16 phase history, intermediates and final-stage operands, float32 accumulation in the filters (float16 build, as released)	-0.26	-0.07
	Release 0.1.0, host memory to host memory	3.48	3.78
CPU	JAX program: conv. filters, direct ramps, one process	-	224.45
	C++ with OpenMP: polyphase filter vectorized along the output columns (stride-D gathers)	-	-
	Children in the vector lanes (no gathers), input rows read once in the pulse filter, final tables by two recurrences	-	-32.02
	Named accumulators, complex products as FMA pairs, 512-bit vectors, reused intermediates	-	-6.58
	Final-stage tables built and consumed three pulses at a time (second-level cache); released build	-	-0.88
	Release 0.1.0, host memory to host memory	-	11.31

E Polar format on an orbital aperture

FastSAR’s polar format consists of three resampling steps and a two-dimensional FFT, each expressed with FFTs, element-wise products or a single matrix product so that the same code runs on all devices. The pulses are first resampled so that the tangent of the look angle in the image plane is uniform in pulse index, which an orbit does not give exactly. This is a 16-tap Kaiser-windowed sinc interpolation along the pulse axis, applied as a sum over 16 gathered rows on every device (a dense 15,186 by 15,186 matrix product, also implemented, needed more than the v5e’s 16 GB with its intermediates in development). Each pulse is then resampled onto a uniform radial wavenumber grid, and each wavenumber row onto a uniform cross-range grid, by chirp-z transforms [38], which evaluate the band-limited interpolation on a scaled grid with FFTs and chirps; outputs that would wrap around the band are zeroed. The frequency samples are weighted by f_c/f_k before resampling, because backprojection’s uniform weighting of the samples corresponds to a density of $1/k$ in the wavenumber plane (the polar Jacobian) that the uniform spectral weighting of polar format omits; in development, omitting this weighting worsened the error by about 1.5 dB.

Because the grazing angle changes along the aperture, the radial band of each pulse projected onto the image plane drifts from pulse to pulse. The raster covers the union of the bands and so keeps the full spectral support of backprojection, a trapezoid, rather than the inscribed rectangle; for these collections the inscribed rectangle would keep about half the bandwidth in the ground plane and nearly all of it in the vendor’s slant plane. The wavenumber grid spacings are set by the native pixel spacings and FFT sizes of 13,824 by 10,800, which include a 300 m guard band on every side of the scene. The range and azimuth profiles are tapered to zero within the guard, because the periodic transforms otherwise wrap echoes from beyond the image into its edges (without the guard the edge error was -5 dB in development).

Distortion. Polar format assumes a planar wavefront. A scatterer at distance ρ from the scene center is then focused about $\rho^2/2R$ from its position in range, which at $R = 744$ km (mid-aperture) is 4 m at 2.5 km, about twelve pixels; the azimuth displacement is larger and depends on the squint (7.6 m in azimuth against 4.3 m in range at the Panama corners). The planar model writes the range history of a scatterer at $\mathbf{p}$ as $-\hat{\mathbf{u}}_p \cdot \mathbf{q} + c$; a least-squares fit of $\mathbf{q}$ and c over the aperture to the exact history $\|\mathbf{p} - \mathbf{a}_p\| - \|\mathbf{a}_p\|$ gives the position at which polar format focuses the scatterer, $(q_x, q_y + c)$, since a constant range offset is a range shift. The fit residual is below 1 mrad of phase everywhere in the three scenes, so the image is displaced without defocus, by up to 18 pixels in azimuth and 12 in range at the Panama corners; in development, measured target positions agreed with the model to 0.2 pixel, and before the final resampling the 5 by 5 coherence with the reference was 0.42. The mapping is fitted by cubic polynomials over the image (residual below 10^{-4} pixel).

Final resampling. The spectrum is placed in a raster of twice the pixel density, transformed without the carrier terms, interpolated at the apparent positions with a six-tap Lanczos kernel along each axis, and multiplied by the carriers evaluated at the apparent positions. No separate phase correction is needed; the quadratic phase of about 12 radians at 211 m seen in development in a center crop before the resampling is the carrier of the range displacement. The oversampled raster is processed in blocks: the range transform is done once, and the azimuth transform and the resampling proceed in blocks of 512 output range columns. Every polar-format figure in this paper is for the resampled image.

Residual. After the resampling the error is -32.1 dB on Panama (-33.5 dB on Melbourne and -35.4 dB on Iowa) and the mean 5 by 5 coherence 0.9994. On every collection the residual grows

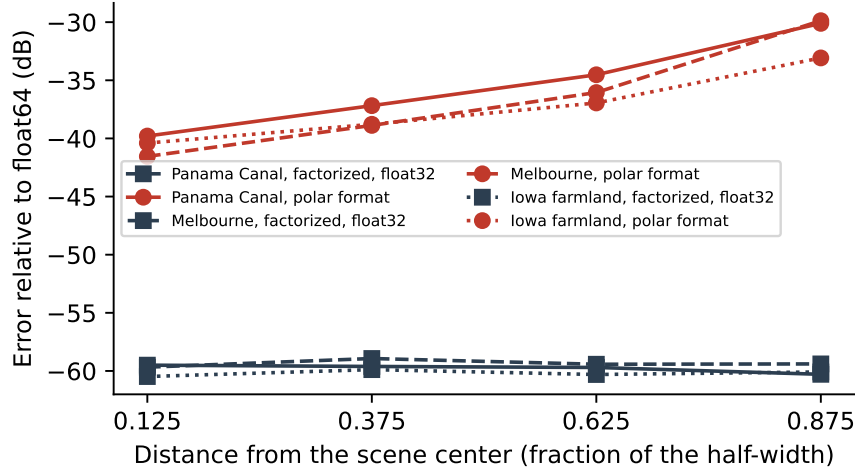


Figure 5: Error relative to the float64 reference in four concentric rings by distance from the scene center, for the polar-format image and the float32 factorized image on the L4, on the three collections (dB; rings are quarters of the half-width from the scene center outward).

Table 10: Panama Canal, polar format on the CPU in development: whole-image error and error by ring (dB) for variations of the final resampling. Rings are the quarters of the half-width from the scene center outward, inner to outer; the selected configuration is the first row.

Variation	Error	Ring 1	Ring 2	Ring 3	Ring 4
6-tap kernel, 2 times oversampled (selected)	-32.1	-39.8	-37.2	-34.5	-30.1
4-tap kernel	-31.6	-37.1	-35.1	-33.5	-29.8
8-tap kernel	-32.2	-40.1	-37.4	-34.7	-30.2
3 times oversampled	-32.2	-40.1	-37.3	-34.7	-30.1
32-tap pulse interpolator, Kaiser 12	-31.7	-37.7	-35.7	-33.8	-29.8
48-tap pulse interpolator, Kaiser 12	-31.2	-36.4	-34.5	-33.1	-29.6

from the inner to the outer ring, by 7 to 12 dB, from -40 to -30 dB on Panama, whereas the factorized error shows no dependence on distance from the center (Figure 5); the cause of the residual was not identified. In development, on the full Panama image, the resampling kernel (4, 6 and 8 taps) and the oversampling of the intermediate raster (2 and 3 times) change the whole-image error by less than 0.6 dB and the outermost ring, where the displacement is 10 to 18 pixels, by less than 0.4 dB (Table 10); lengthening the pulse interpolator from 16 to 32 or 48 taps, with a narrower Kaiser window, made the error up to 0.9 dB worse. The registration error, measured in development from the cross-spectrum phase, was 0.001 pixel, and the convergence of the reference is more than 30 dB below the residual. One untested candidate is the Taylor weighting, which backprojection applies per pulse and sample and polar format applies through the resampling onto the rectangular raster; such a difference would grow with distance from the scene center.

Time. The final resampling is an indexed read at a computed position, 36 per pixel. On Panama the polar-format image takes 3.0 s on the L4 and 11.4 s on the CPU, 0.80 and 1.01 times the factorized float32 time on the same device. On the TPUs it takes 103.4 s (v5e) and 110.2 s (v6e), much slower

Table 11: The reference (float64 exact backprojection) against the vendor’s complex image (SICD) of the same collection, which Umbra forms by polar format. Columns: the number of 512 by 512 windows, centered on a grid with 1024-pixel spacing, with enough contrast for an amplitude cross-correlation; the largest displacement of the vendor image relative to the reference over the windows (azimuth / range, pixels); the largest displacement the planar-wavefront model of Appendix E predicts at the same windows; the root-mean-square difference between measured and predicted displacement; the root-mean-square residual displacement after the vendor image is resampled through the model; and, after that resampling and a local re-alignment and phase-ramp removal in each window, the 5 by 5 coherence of the two images (mean over all pixels of the scene; median of the window means) and the median over the windows of the correlation of their log amplitudes.

Collection	Windows	Measured (px)	Model (px)	RMS difference (px)	Residual after resampling (px)	Coherence, scene mean	window median	Log-amplitude correlation
Panama Canal	68	13.2 / 9.0	13.2 / 9.1	0.11 / 0.12	0.07 / 0.05	0.46	0.67	0.49
Melbourne	81	7.3 / 7.1	7.4 / 7.2	0.10 / 0.19	0.05 / 0.11	0.67	0.76	0.52
Iowa farmland	64	6.7 / 8.1	6.7 / 8.5	0.07 / 0.34	0.01 / 0.25	0.63	0.80	0.57

than factorized backprojection. Its final resampling consists of scattered two-dimensional gathers, at the rate given in Appendix C, whereas the contiguous row gathers of the pulse resampling delivered nearly a hundred times more elements per second in the same microbenchmarks. A separable approximation of the final resampling, expressible as row and column phase ramps, would avoid the scattered gathers, but the squinted geometry of these collections is not exactly separable in this way, and its effect on the TPU’s cost was not measured.

F Validation against the vendor’s image

Every error figure in this paper, except the ISCE3 scores against its own receive model (Section 5.4), is measured against the reference, the float64 exact backprojection of Section 4, implemented for this work. Umbra, the vendor, also distributes each collection as a complex image in the Sensor Independent Complex Data format (SICD), formed from the same phase history by its production processor using polar format, on the pixel grid that the reference adopts; this allows the reference to be checked against an independent processor. The integer and sub-pixel offset between the two images (attributed to the vendor’s reference-pixel convention) was found from the cross-correlation of block-averaged log amplitude and refined on a central window; the linear phase ramp between them, which reflects the different spectral centering of the two images (-0.225 cycles per pixel in range on Panama), was found from the peak of the cross-spectrum of the same window and removed. The vendor image was compared with the reference in 512 by 512 windows centered on a grid with 1024-pixel spacing, keeping the windows with enough contrast for an amplitude cross-correlation (68 on Panama). In each window the local offset of the vendor image was measured by amplitude cross-correlation and compared with the displacement that the planar-wavefront model of Appendix E predicts at the window center. The vendor image was then resampled through the model with a cubic kernel and, after a local re-alignment and phase-ramp removal in each window, the coherence and the log-amplitude correlation of the two images were computed. As delivered, the vendor image differs visibly from the reference in two respects: it is masked to the parallelogram over which the vendor’s processor has full spectral support, and bright points carry cross-shaped sidelobes that the reference’s Taylor weighting suppresses.

Table 11 gives the results. The vendor image is displaced from the reference by a field whose largest value over the Panama windows is 13.2 pixels in azimuth and 9.0 in range, against 13.2 and 9.1 predicted by the model from the collection geometry alone. The root-mean-square difference

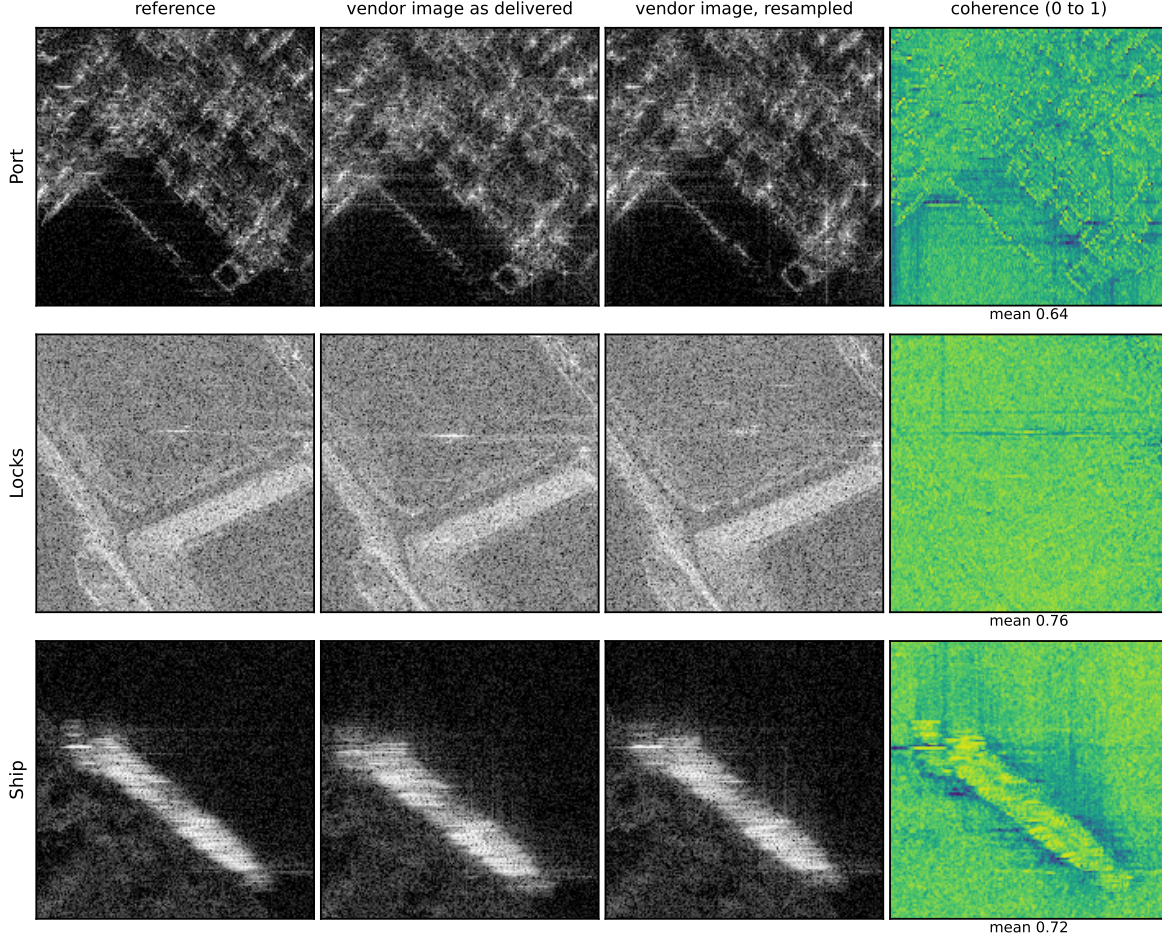


Figure 6: The port, lock and ship regions of Figure 3 (rows, 512 by 512 pixels). Columns: the reference; the vendor image as delivered, on the same pixels; the vendor image after resampling through the planar-wavefront displacement model of Appendix E; and the 5 by 5 coherence of the resampled vendor image with the reference after local re-alignment and phase-ramp removal, from 0 to 1, with its mean below each map. Amplitudes are displayed over a 45 dB range.

between measured and predicted displacement over the windows is 0.11 and 0.12 pixel on Panama and at most 0.34 pixel on the other two collections, 1 to 4% of the largest displacement. After the resampling the root-mean-square residual displacement is at most 0.25 pixel (individual windows up to 0.43). The coherence of the two images averages 0.46, 0.67 and 0.63 on Panama, Melbourne and Iowa (0.67, 0.76 and 0.80 as the median over windows), and their log amplitudes correlate at 0.49 to 0.57 within windows (Figure 6). At the brightest isolated points of the lock and port regions (the ship is a moving target and the vegetated region has no isolated point) the impulse-response widths of the two images differ by 1 to 27%. The vendor's response is wider in azimuth at both points and, in range, narrower at one and 1.3% wider at the other, which a different aperture weighting in the vendor's processor would produce. The SICD metadata does not state whether autofocus was applied, and the vendor's motion compensation is independent of the one used here. Coherence at this level is attributed mainly to the differences between the two processors and is not used as a measure of precision. The vendor image is delivered without a correction for the displacement; the polar-format images of this study remove it in the final resampling (Appendix E).

Table 12: Polar-format implementations on the Panama collection, on the CPU instance: correlation of the log amplitude with the reference and mean coherence in 5 by 5 pixel windows over the three regions, and the time per full image.

Implementation	Log-amplitude correlation	Coherence	Time per full image
AFRL toolbox <code>pfa_mem</code> (Octave)	0.80 to 0.89	0.76 to 0.88	44 s
GRDL <code>PolarFormatAlgorithm</code>	0.78 to 0.93	0.83 to 0.94	24.5 s
FastSAR polar format, float32	0.996 to 0.999	0.999	11.4 s

G Comparison with open-source implementations

RITSAR [17] (commit 0e36d2e, ported to Python 3) is a NumPy library, and `bpBasic` is the backprojection of the MATLAB toolbox of Gorham and Moore [2], written at the Air Force Research Laboratory and distributed by the National Geospatial-Intelligence Agency (commit 63f0401), run in GNU Octave 8.4. ISCE3 [19] (0.25.17; 0.25.21 for the Capella sliding spotlight) has C++/OpenMP and CUDA versions, and `torchbp` [20] (commit b371884) is a PyTorch extension with CUDA kernels. GRDL [18] (commit de3338a) is a Python library with Numba kernels that reads CPHD files and implements polar format and fast factorized backprojection, run with its default parameters. Each received the Taylor-windowed phase history and antenna positions of the reference (Appendix A) in its input format and was scored with the error measure of Section 4 on three of the regions of Figure 3, ISCE3 on 256 by 256 pixel patches of its own radar grid. Each had the whole instance; ISCE3 kept about 5.6 of the 16 vCPUs busy. The scripts, records and a longer account of each comparison are in the directory `results/oss` of the study repository.

RITSAR’s error arises from two indexing defects, a range axis built with `linspace` whose spacing is too large by one part in $N_{\text{fft}} - 1$ and a frequency axis centered on the assumption of an even sample count, which the 14,399 samples of the collection do not satisfy. The error of `bpBasic` falls by 12 to 15 dB per doubling of its FFT length, as linear interpolation of the range profiles predicts, but at 2^{17} stays 15 to 17 dB above FastSAR’s exact backprojection with linear interpolation of profiles oversampled 8 times (region errors in the study records), which is attributed to its referencing the phase of its profiles to the lowest frequency of the band. ISCE3’s amplitude correlates with the reference at 0.999 or more; most of its complex difference is a linear phase ramp common to all patches. A difference in spectral centering produces such a ramp. Once the ramp is removed, the center patch agrees to -54.9 dB and the off-center patches retain an unexplained residual near -32 dB. `Torchbp` evaluates the distance in float32 (Table 3, footnote). Its exact backprojection forms no image, whereas a float64 evaluation of the same formula does. Its factorized backprojection required more than 23 GiB, which exceeds the 22.0 GiB that PyTorch reports as the L4’s capacity. GRDL’s factorized backprojection, which its documentation describes as a stripmap algorithm, ran on the CPU in 77.5 s but its image of the spotlight collection is smeared in cross-range, and no orientation of its grid registers to the reference (amplitude correlation 0.22 or less).

The AFRL toolbox also contains a polar format, `pfa_mem`, run unmodified in Octave on the same instance (with a compatibility version of `cross` for mixed row and column inputs), and GRDL’s polar format ran on the same instance. Neither image is corrected for the planar-wavefront displacement, so each region was registered and resampled onto the reference pixels before scoring by amplitude statistics and coherence, as for the vendor image of Appendix F (Table 12); part of their coherence loss comes from that resampling.

H Measurement protocol in detail

Time and cost. The software comprised FastSAR release 0.1.0 or the release candidates below, JAX 0.11.2 [34], CUDA 12 with CuPy 14.2.0 [36], and Numba 0.68.0 [30] for the reference. The Umbra records were made at the release-candidate tags `rel-rc3-20261009` (CPU and TPUs) and `rel-rc5-20261009` (L4), the Capella records at `rel-rc2-20261009` (CPU and v6e) and `rel-rc5-20261009` (L4); at `rel-rc2` and `rel-rc3` the library code differs from release 0.1.0 only in the CUDA path and the GPU memory settings, which the CPU and TPU records do not exercise, and at `rel-rc5` only in one docstring. Each Umbra record names its commit, and each Capella record’s log does. Each configuration ran in its own process through the public interface. Under the protocol of Section 4 the GPU and TPU times include moving the collection to the device and the image back, and reading the collection from disk is excluded everywhere. For the factorized algorithm the interval also includes the per-collection host work (the float64 geometry of all three levels for the CUDA and C++ kernels and of the first two for the TPU kernels). Each configuration was run once to compile and then twice more, and the shorter of the two is reported; the first warm call was up to 32% slower where one-time work fell on it (90th percentile of the difference 5%), and on the CPU and the L4 the reported time agrees with the loop’s time per image within 3%. Differences of a few percent between rows are not resolved.

The loop ran until at least four images had been formed and at least 60 s had elapsed; on the CPU, exact backprojection and the JAX program were timed without a loop. On the TPUs, staging overlaps formation as described in Section 4, through `ImageFormer.stage`, as in the mosaics. Throughput is the number of images divided by the elapsed time, and the cost per thousand images is computed from it.

The process’s CPU time was recorded during every run. The CPU energy estimate is described with Table 13.

Power. The L4’s draw was sampled from `nvidia-smi` twice a second during the timed loops. The TPU virtual machines and the CPU instance expose no power counters; the TPU ranges, derived from third-party design-power figures, and the pro-rata CPU estimate are in Table 13 and are marked as estimates.

Precision. The gain fitted per algorithm family is a complex gain for the backprojections and a real one for polar format, whose phase differs from the reference by a smooth screen before the final resampling. Six statistics are computed for each image. The error is the energy of the difference relative to the energy of the reference (dB); this is the figure quoted when a single number is given. The others are the largest single-pixel difference relative to the brightest pixel (dB); the complex coherence between the two images over a 5 by 5 window, as the mean, the 0.1 percentile and the minimum; the amplitude ratio and the phase difference over the brighter half of the pixels, as the 99th percentile and the maximum; and the error in four rings by distance from the scene center. Table 14 gives the main statistics of the Panama configurations, and the supplementary tables give all six for every configuration on every collection.

CPU details. On the CPU the factorized float32 row comes from the C++ kernels of Appendix D and exact backprojection from its C++ kernel; polar format runs as a JAX program. The reference was formed on a separate n2-highmem-48 instance (48 vCPUs, \$3.14 per hour on demand) in 1892 s per Panama image (about \$1,652 per thousand images) and is not part of the cost comparison.

Table 13: Energy per Panama image for the processor alone, over the single-image time (the kWh column of Table 4 uses the per-image time of the timed loop, which is shorter than the single-image time on the TPUs only). The L4 draw is the mean of `nvidia-smi` samples during the timed loop; the TPU figures are ranges based on design-power figures quoted by third parties (120 to 200 W for the v5e, 200 to 350 W for the v6e [39, 40]), since the announcement of the v6e [41] gives no power figure, assuming the chip ran at that power throughout; the CPU figure is the pro-rata share of the 8 cores (25 W of the 400 W package) times their occupancy, the busy fraction of the 16 hardware threads doubled and capped at one. Hosts, memory and the other components of each instance are excluded.

Device	Configuration	Basis	Power (W)	Time (s)	Energy (kJ)
L4	Exact BP, cubic interpolation, $4\times$ oversampled	measured	71	18.7	1.33
L4	Factorized BP, float16 throughout, CUDA kernels	measured	60	3.5	0.21
L4	Factorized BP, float32, CUDA kernels	measured	64	3.8	0.24
TPU v5e	Factorized BP, single-pass products, fused kernels	third-party design power	120 to 200	3.6	0.4 to 0.7
TPU v5e	Factorized BP, three-pass products, fused kernels	third-party design power	120 to 200	5.2	0.6 to 1.0
TPU v6e	Factorized BP, single-pass products, fused kernels	third-party design power	200 to 350	2.6	0.5 to 0.9
TPU v6e	Factorized BP, three-pass products, fused kernels	third-party design power	200 to 350	3.1	0.6 to 1.1
CPU	Factorized BP, float32, C++ kernels	pro-rata estimate	25 at 100%	11.3	0.28

Reference convergence and exact backprojection. The reference is a float64 exact backprojection with range profiles oversampled 16 times. Its convergence was measured on a 1024 by 1024 crop by forming the same crop at 8, 16 and 32 times: the 8 times image differs from the 32 times image by -55.5 dB and the 16 times image from the 32 times image by -69.2 dB. At 8 times the convergence error would have been comparable to the differences under study, so the 16 times reference was used. This sets the resolution limit stated in Section 4.

FastSAR’s exact backprojection with linear interpolation of 16 times oversampled profiles, the interpolation of the reference, measures -80.5 dB against it; the shared interpolation error cancels, and the figure indicates only agreement between the two implementations. With linear interpolation the error is set by the profile oversampling, -56.9 dB at 8 times. Cubic interpolation at 4 times oversampling, the default, measures -70.0 dB, at the limit to which the reference itself has converged, so its true error may be lower. With 16 times oversampling the L4 takes 26.9 s per image, \$5.3 per thousand images.

Limitations of the cost comparison. The improvements still identified in the factorized kernels are about 12% on the TPUs and 8% on the L4 in the profiled builds, and none was quantified on the CPU, whose stages run at 1.05 to 1.4 times their bounds (Appendix D). The factorized plan (32-pixel final tiles, three levels, a passband fraction of 0.4 and 70 dB filter attenuation) kept its defaults on every device; a different factorization would change the bounds themselves, by an amount not measured here. TPU7x and H100 instances could not be obtained under the project’s cloud quota and were not evaluated. The cost of the asterisked row of Table 4 is computed from the single-image time, not from a loop. The TPU hosts (24 and 44 vCPUs) are larger than the work needs, while the L4’s (4 vCPUs) is the smallest offered. Only on-demand prices were used; spot and committed-use prices were not considered.

I Ground truth of the Panama regions

The identity of the Panama regions used in Figure 3 was checked against Copernicus Sentinel-2 imagery [28]. Each pixel of the images formed here was located on the ground from the vendor’s SICD header: the pixel’s position in the slant plane follows from the scene center point and the row and column unit vectors, and the ground point is the intersection of the line through it along the slant plane’s normal with the WGS84 ellipsoid at the scene height. The image corners so computed agree with the corner coordinates in the SICD header to within 15 m. The nearest usable Sentinel-2 acquisition is from 2023-07-23, five days later, at 10 m resolution; a dry-season acquisition of 2024-02-23 is shown as well because clouds cover part of the earlier one. The lock chambers, the quays and container yard of the terminal, and the road beside the vegetation region coincide in the two sensors. The bright target of the ship region, about 180 m long and close to the eastern bank, does not appear in the vendor’s geocoded products of six other passes of this repeat-collection site, four of them three, four, five and seven days after this collection and two in February 2024. The pass of 2023-07-25 02:30 UTC repeats the geometry of this collection to within 0.3° (azimuth 122.4° against 122.1° , grazing angle 42.6° in both, left-looking), which makes an aspect-dependent return from a fixed object unlikely; the other passes span azimuths from -72° to 223° and grazing angles from 40° to 63° . The pass of 2023-07-23 14:47 UTC precedes the Sentinel-2 image of that day by 64 minutes, and neither shows anything at the position. It is taken to be a ship in the navigation channel; the Sentinel-2 image of 2024-02-23 shows a different ship nearby. The vendor’s geocoded product of this collection places the target inside the outlined crop as well.

The ship appears on the eastern bank, inland of the channel, as a consequence of its motion during the aperture. A stationary-scene processor places a target with line-of-sight velocity v_r at the position whose Doppler history matches, displaced along azimuth by about Rv_r/V (here about 100 m per m/s), and the along-track velocity defocuses it [42]. Since exact backprojection takes the antenna position of every pulse as data, a target moving at constant velocity $\mathbf{v}$ is formed exactly by backprojecting with the track shifted by $-\mathbf{v}t$, which both focuses it and places it at its position at the aperture center. The ship was assumed to follow the channel axis (bearing 158° , from the Sentinel-2 water mask) and its speed was swept from -8 to 8 m/s in a tile around it (Figure 7). The sharpness of the tile, the sum of the squared intensities, is single-peaked at 3.0 m/s (5.8 knots toward the Pacific), where the peak intensity is 10.6 dB above the stationary-scene image and the ship has moved 469 pixels (194 m) along azimuth into the navigation channel; the position of the water was not used in the estimate. A heading 10° farther east (148°) gives the same speed with an 18% lower sharpness peak and a compensated position 22 m away. The images compared in Figures 3 and 4 are the stationary-scene images, in which the ship appears at the same displaced position under every algorithm; the motion analysis was used only to identify the target.

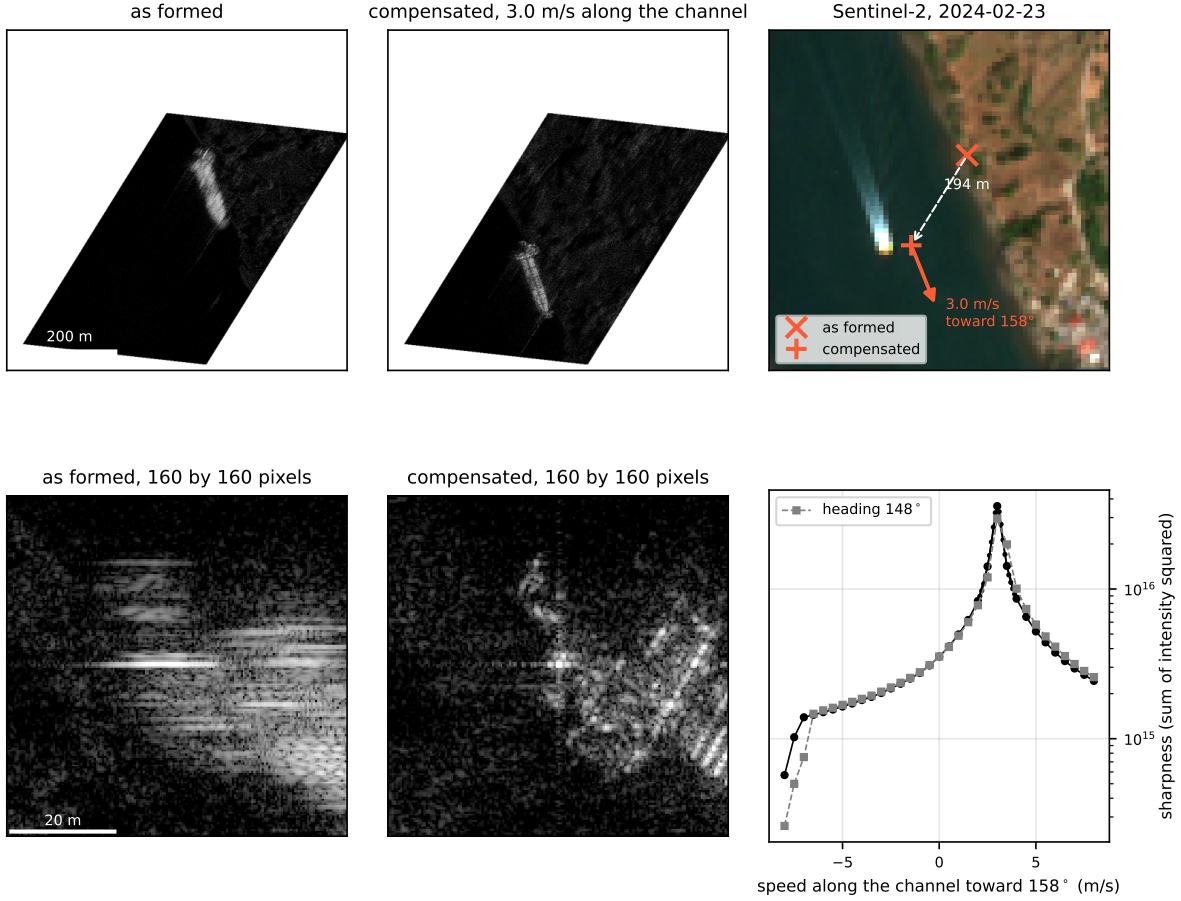


Figure 7: The ship compensated for its motion. Top: the tile around the ship as formed and after backprojection with the antenna track shifted for a ship moving at 3.0 m/s along the channel, both resampled north up over a 760 m window (200 m bar), and the two positions on the Sentinel-2 image of 2024-02-23, with the 194 m displacement (dashed) and the estimated velocity, 3.0 m/s toward 158°, drawn at the compensated position. Bottom: 160 by 160 pixel windows around the ship in image coordinates (20 m bar), and the sharpness of the tile against the assumed speed for the channel heading of 158° (coarse and fine sweeps) and for 148°.

Table 14: Precision of the Panama Canal images against the float64 exact backprojection: energy of the difference (dB), largest pixel difference relative to the brightest pixel (dB), 0.1 percentile and minimum of the 5 by 5 coherence (the mean is in the supplementary tables), amplitude ratio (99th percentile and maximum, dB) and phase difference (99th percentile and maximum, degrees) over the brighter half of the pixels. The 16 times oversampled exact row uses the interpolation of the reference (Appendix H).

Device	Configuration	Error	Max	Coh. p0.1	Coh. min	Amp. p99	max	Phase p99	max
L4	Exact BP, cubic interpolation, 4× oversampled	−70.0	−79.5	1.0000	0.998	0.01	0.7	0.1	5
CPU	Exact BP, cubic interpolation, 4× oversampled	−70.0	−79.5	1.0000	0.998	0.01	0.7	0.1	5
L4	Exact BP, linear interpolation, 8× oversampled	−56.9	−62.2	0.9998	0.923	0.03	2.5	0.2	15
CPU	Exact BP, linear interpolation, 8× oversampled	−56.9	−62.2	0.9998	0.923	0.03	2.5	0.2	15
L4	Exact BP, linear interpolation, 16× oversampled	−80.5	−88.8	1.0000	1.000	0.00	0.1	0.0	0
L4	Factorized BP, float32, CUDA kernels	−59.9	−65.0	0.9999	0.931	0.02	4.2	0.1	20
L4	Factorized BP, float16 throughout, CUDA kernels	−58.9	−64.7	0.9999	0.932	0.03	4.2	0.2	20
CPU	Factorized BP, float32, C++ kernels	−60.0	−65.0	0.9999	0.931	0.02	4.2	0.1	20
CPU	Factorized BP, float32, JAX program	−58.3	−64.8	0.9999	0.931	0.03	4.2	0.2	20
TPU v6e	Factorized BP, three-pass products, fused kernels	−57.8	−64.5	0.9999	0.931	0.03	4.2	0.2	20
TPU v6e	Factorized BP, single-pass products, fused kernels	−46.2	−64.8	0.9981	0.674	0.16	10.4	1.1	106
TPU v5e	Factorized BP, three-pass products, fused kernels	−57.9	−64.8	0.9999	0.931	0.03	4.2	0.2	20
TPU v5e	Factorized BP, single-pass products, fused kernels	−46.2	−65.0	0.9981	0.674	0.16	10.9	1.1	106
L4	Polar format, float32, gather	−32.1	−40.3	0.9857	0.680	0.69	20.7	6.1	70
TPU v6e	Polar format, float32, gather	−32.1	−40.3	0.9857	0.680	0.69	20.7	6.1	70
TPU v5e	Polar format, float32, gather	−32.1	−40.3	0.9857	0.680	0.69	20.7	6.1	70
CPU	Polar format, float32, gather	−32.1	−40.2	0.9857	0.679	0.69	20.1	6.1	70

J Precision and cost tables

Table 14 gives the error statistics of the Panama configurations and Table 15 the cost figures for Melbourne and Iowa; the time and statistics of every configuration on each scene are in the supplementary tables of the study repository ([report/v2/supplement_tables.pdf](#)). Configurations are labeled as in the body: on the TPUs *single-pass* and *three-pass products* are the one and three bfloat16 passes of a float32 product; *float16 throughout* is the L4’s float16 build; *JAX program* marks the pure JAX implementation of Appendix D; exact backprojection is labeled by its interpolation and the oversampling of its range profiles.

Table 15: Melbourne and Iowa: time, throughput and cost for the configurations of Table 4 except exact backprojection on the CPU; bold marks the fastest configuration on each collection (in Table 4 bold marks the Pareto front).

Device	Algorithm	Arithmetic	Time (s)	Images/h	\$ per 1000	kWh per 1000
<i>Melbourne</i>						
L4	Factorized BP	float16 throughout, CUDA kernels	3.6	1,004	0.70	0.06
L4	Factorized BP	float32, CUDA kernels	3.9	901	0.78	0.07
L4	Polar format	float32, gather	3.1	1,156	0.61	0.04
L4	Exact BP	cubic interpolation, 4× oversampled	19.1	187	3.77	0.38
L4	Exact BP	linear interpolation, 8× oversampled	20.6	178	3.98	0.40
TPU v5e	Factorized BP	single-pass products, fused kernels	3.7	1,462	0.82	0.08 to 0.14
TPU v5e	Factorized BP	three-pass products, fused kernels	5.5	863	1.39	0.14 to 0.23
TPU v5e	Polar format	float32, gather	68.2	53	22.75	2.28 to 3.79
TPU v6e	Factorized BP	single-pass products, fused kernels	2.6	2,377	1.14	0.08 to 0.15
TPU v6e	Factorized BP	three-pass products, fused kernels	3.1	1,791	1.51	0.11 to 0.20
TPU v6e	Polar format	float32, gather	106.5	34	79.87	5.92 to 10.35
CPU	Factorized BP	float32, C++ kernels	12.8	285	3.38	0.09
CPU	Polar format	float32, gather	11.9	299	3.23	0.08
<i>Iowa farmland</i>						
L4	Factorized BP	float16 throughout, CUDA kernels	2.3	1,578	0.45	0.04
L4	Factorized BP	float32, CUDA kernels	2.5	1,418	0.50	0.05
L4	Polar format	float32, gather	2.3	1,595	0.44	0.03
L4	Exact BP	cubic interpolation, 4× oversampled	11.6	311	2.27	0.23
L4	Exact BP	linear interpolation, 8× oversampled	12.6	281	2.52	0.25
TPU v5e	Factorized BP	single-pass products, fused kernels	2.5	2,173	0.55	0.06 to 0.09
TPU v5e	Factorized BP	three-pass products, fused kernels	3.5	1,342	0.89	0.09 to 0.15
TPU v5e	Polar format	float32, gather	48.3	75	16.10	1.61 to 2.68
TPU v6e	Factorized BP	single-pass products, fused kernels	1.8	3,329	0.81	0.06 to 0.11
TPU v6e	Factorized BP	three-pass products, fused kernels	2.2	2,538	1.06	0.08 to 0.14
TPU v6e	Polar format	float32, gather	61.1	59	45.85	3.40 to 5.94
CPU	Factorized BP	float32, C++ kernels	7.3	477	2.02	0.05
CPU	Polar format	float32, gather	8.4	424	2.27	0.06